

Buffered comparison and stopping-band resolution in critical Ising

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Abstract

We prove pointwise comparison and finite stopping-band approximation theorems for the critical nearest-neighbor Ising model on the square lattice. Arbitrary common pinned spins are allowed in the comparison: the likelihood ratio is controlled by a zero-field FK connection probability on the graph with those vertices deleted. For an interface exploration across a band of fixed positive width, finitely many regular signed barriers approximate every good conditional target law in total variation, uniformly over bounded observables and sufficiently fine meshes.

1 Introduction

An exploration of an Ising interface fixes spins along a random boundary. To use the unexplored region, one needs estimates that remain valid after this conditioning. Two difficulties arise. The fixed signs can alternate, and the exposed boundary can contain many narrow excursions. We give comparison estimates that retain arbitrary common signs, then replace the boundary visible from a distant target by one of finitely many regular signed barriers. The replacement controls the whole target spin law, including observables that vary with the mesh.

1.1 Finite graphs and pointwise comparison

For a finite graph $G = (V, E)$, nonnegative couplings K_e , and finite real fields h_v , the Ising law is

$$\mu(\sigma) = Z^{-1} \exp \left(\sum_{\{u,v\} \in E} K_{uv} \sigma_u \sigma_v + \sum_{v \in V} h_v \sigma_v \right), \quad \sigma \in \{-1, +1\}^V.$$

Prescribing a spin removes that variable and adds its incident interactions to the fields of its neighbors. By contrast, deleting a vertex means removing the vertex and all its incident edges.

Let I, J, F be disjoint, with I, J nonempty, and prescribe common spins on F . Form a comparison graph by deleting F , contracting I to one terminal and J to another, and setting all fields to zero. Let q_0 be the probability that the two terminals are connected in its random-cluster model with cluster weight 2 and edge parameters $p_e = 1 - e^{-2K_e}$. Parallel couplings are added on contraction. Write $\|\nu - \nu'\|_{\text{TV}} = \sup_A |\nu(A) - \nu'(A)|$.

Theorem 1.1 (Finite-graph comparison). *For every two configurations b, b' on J and every configuration a on I ,*

$$e^{-4 \operatorname{arctanh} q_0} \leq \frac{\mu(\sigma_I = a \mid \sigma_J = b, \sigma_F)}{\mu(\sigma_I = a \mid \sigma_J = b', \sigma_F)} \leq e^{4 \operatorname{arctanh} q_0}.$$

Either conditional law may be replaced by an arbitrary mixture of these laws. The bound is symmetric in the two terminal sets.

Lemma 2.2 proves the theorem. For the square lattice at

$$\beta_c = \frac{1}{2} \log(1 + \sqrt{2}), \quad B_r = [-r, r]^2 \cap \mathbb{Z}^2,$$

dual crossings turn it into a uniform buffered comparison. If $I \subset z + B_R$ and $J \subset (z + B_{(1+c)R})^c$, the density ratio is bounded above and below by constants depending only on $c > 0$. If $I \subset z + B_w$, $J \subset (z + B_R)^c$, and $R \geq 8w$, its logarithm is at most $C(w/R)^\alpha$ in absolute value, for an absolute $\alpha > 0$. Common pins may occur anywhere off the terminals. The comparison also passes through a common conditional randomization, such as internal Edwards–Sokal bond sampling. Precise geometric and cross-domain forms appear in Corollaries 2.5 and 2.6.

1.2 Exploration across a band

For the exploration results we use the critical nearest-neighbor Ising model with zero external field and prescribed boundary spins. Put $Q_r = \{z : \|z\|_\infty < r\}$. Fix $0 < u < v$, a larger square Q_L , and a target at positive distance from the band $\bar{Q}_v \setminus Q_u$ and the boundary of Q_L . In the inward construction, reveal the outside of Q_v and explore towards Q_u ; the target is a compact subset of Q_u . In the outward construction, reveal Q_u and explore towards Q_v ; the target lies between ∂Q_v and a more distant fixed wall. All the clearances are fixed before the lattice mesh δ tends to zero. They can be arbitrarily small positive numbers. When the two contour levels vary, each ranges over a prescribed open interval. Their closures are disjoint and preserve all these clearances; these are the separated level windows.

Choose a fixed diagonal in every lattice face and linearly interpolate the spins on the resulting triangles. The zero set gives noncrossing polygonal strands. Explore each strand meeting the starting contour until its first return to that contour or its first hit of the other contour, revealing the traversed triangles' spins. Include the starting contour vertices in the initial reveal. The complete transcript is the record of this adaptive reveal. The Gibbs property holds conditional on each transcript, as proved in Lemma 7.2.

A *regular signed barrier* is a deterministic Jordan contour with finitely many prescribed plus and minus arcs, together with a lattice boundary layer following those arcs. Its fresh side is the side containing the target. A finite descriptor is a function of the band spins with values in a fixed finite set; one value is allowed to be exceptional.

Theorem 1.2 (Finite conditional resolution). *Fix the preceding band geometry and use the critical plus-boundary law in a larger reference square. For every $\epsilon, \eta > 0$, one can choose contour levels in any prescribed separated subwindows, a finite descriptor $Z^\delta \in \{0, 1, \dots, m\}$, and regular signed barriers $\Gamma_1, \dots, \Gamma_m$ with the following properties for all sufficiently fine meshes.*

The exceptional state 0 has probability at most ϵ . For every transcript in a good state $i \in \{1, \dots, m\}$, insert the plus pins of Γ_i and then its minus pins. The sum of the two resulting total-variation changes on the target is at most η . The final target-side law $Q_{i,\delta}$ is the same for every transcript in that state. Consequently this conclusion holds simultaneously for all bounded target observables, including mesh-dependent observables. For any fixed finite list of bands the descriptors have convergent joint laws.

The finite list of barriers and its positive geometric clearances are chosen before the mesh threshold. If another law has band density at most K times the reference density, its exceptional probability is at most $K\epsilon$.

The full statement, including a further inward barrier displacement, is Theorem 9.1. Density comparison transfers the probability of the exceptional event. To transfer a conditional target law,

the continuation beyond a common source contour must also be the same; Corollary 9.2 states this distinction precisely.

1.3 Inputs and proof mechanism

Critical spatial comparison already follows from FK crossing estimates. Duminil-Copin, Hongler, and Nolin prove polynomial relative decorrelation for separated FK events and discuss ratio mixing [6, Section 5.3, Proposition 5.11, author-hosted version]. Here we formulate comparison directly for full spin configurations on arbitrary terminal sets of a finite ferromagnetic graph, retaining arbitrary common pins and fields. The positive-association and cluster representation framework is due to Fortuin–Kasteleyn–Ginibre, Holley, and Edwards–Sokal [8, 9, 7]; marginal closure of the lattice condition belongs to the multivariate total positivity theory of Karlin and Rinott [10]. We give the binary marginalization proof and reduce terminal log-odds rectangles to the extreme rectangle. Balancing the two terminal fields then permits the arbitrary-field covariance inequality of Ding, Song, and Sun [5, Corollary 1.3]. That inequality is the imported two-spin result; the pointwise terminal comparison is the finite-graph deduction proved here. The FK crossing and one-arm estimates of [6] supply its critical geometric consequences.

To control a conditional target law, we construct an ordered coupling in a doubled spin model. A disagreement that reaches a patch carries magnetic mass of order $w^{7/8}$ on a nearby counting segment of scale w . Pointwise comparison bounds the effect of changing that patch on a target at distance s . Summing over an admissible cut gives ordered transmission, including its signed-source and total-variation forms (Section 4). As a separate quantitative consequence of the spin-correlation limits of Chelkak, Hongler, and Izyurov [4], we prove a mesh-first L^2 bound of order $(w/s)^{3/8}$ between normalized magnetizations on nearby contours, and a prediction bound from a nearby contour (Section 5). The stopping construction below uses transmission directly.

Benoist and Hongler prove convergence of critical Ising interfaces to nested CLE_3 and a simultaneous common-edge approximation for all macroscopic Ising loops [2]. We use that stronger approximation to preserve coverage, orientation, and multiplicity under every noncrossing complete resolution of the lattice interfaces (Section 6). At generic stopping levels the part of the boundary visible from the target is a Jordan base with finitely many slits (Section 7).

The main geometric step replaces this slit boundary by regular signed outlines. Finite graph embeddings and planar extension theorems [13, 11] provide a topological chart. Narrow ribbons retain the two boundary sides of each slit and the sign switch at its tip. The spin-correlation limits of [4] control magnetic means in fixed regular upper and lower comparison domains. Small neighborhoods of roots and tips can be overwritten with common pins at a vanishing cost on a distant test contour. Away from them, the signs seen by the common free component order every nearby realized wall between the regular comparisons. Ordered transmission then turns the small mean gap into target total variation. For an outward band, a common bounded weight also permits arbitrary prescribed signs on the distant wall.

This construction gives one barrier for every realizable transcript in a neighborhood of the limiting slit datum. Its neighborhood and mesh threshold are chosen before the target observable. A countable cover, reduced to finitely many continuity sets carrying arbitrarily high probability, yields Theorem 1.2 (Sections 8–9).

The positive width of every comparison collar is a fixed geometric hypothesis; the constants may depend on it. Pointwise ratios allow arbitrary common pins, whereas the magnetic mass charts and test contours used in transmission have the free neighborhoods specified in Definition 4.1. These separate hypotheses will be kept explicit throughout.

2 Pointwise comparison through buffers

We bound the probability of every local spin configuration after changing prescribed spins, uniformly over all common pins. We prove a finite-graph estimate in terms of a zero-field FK connection probability, and then use critical crossings to control that probability through lattice collars. All constants in the resulting geometric bounds are independent of the number of lattice vertices.

2.1 A finite-graph bound from extreme log odds

We first allow any finite graph, nonnegative pair couplings, and arbitrary finite fields. On a binary cube, a positive weight p satisfies the *lattice condition*, or is *log-supermodular*, if

$$p(x \wedge y)p(x \vee y) \geq p(x)p(y) \quad \text{for every } x, y,$$

where the minimum and maximum are taken coordinatewise.

The preservation of this condition under marginalization is a basic property of multivariate total positivity [10]. We give the finite binary argument and then show that every log-odds rectangle is bounded by the rectangle between the four constant terminal configurations. This reduces the likelihood-ratio problem to two spins.

Lemma 2.1 (Extreme rectangles in a ferromagnetic marginal). *Let $p(a, b) > 0$ be a marginal of a finite ferromagnetic Ising law with arbitrary fields, on two disjoint sets of spins. Set $f = \log p$ and*

$$L = f(+, +) + f(-, -) - f(+, -) - f(-, +).$$

For every four configurations a, a', b, b' ,

$$|f(a, b) + f(a', b') - f(a, b') - f(a', b)| \leq L. \quad (2.1)$$

Proof. The Ising weight is log-supermodular: each ferromagnetic pair interaction has nonnegative mixed difference and each field has zero mixed difference. We recall why summing out spins preserves this property. It suffices to check a two-coordinate square after summing one other binary coordinate. Denote the eight unsummed weights by a, b, c, d, e, f, g, h , where (a, b, c, d) and (e, f, g, h) are the two square slices, in order $(--, +-, -+, ++)$. Log-supermodularity gives $ad \geq bc$, $eh \geq fg$, $ah \geq bg$, and $ah \geq cf$. Moreover $(ah)(ed) = (ad)(eh) \geq (bg)(cf)$. For positive numbers X, Y, U, V , the conditions $X \geq \max(U, V)$ and $XY \geq UV$ imply $X + Y \geq U + V$: assume $U \geq V$ and use $X + UV/X \geq U + V$ for $X \geq U$. Applying this with $(X, Y, U, V) = (ah, ed, bg, cf)$ gives $ah + ed \geq bg + cf$. Hence $(a + e)(d + h) \geq (b + f)(c + g)$. Nonnegative adjacent-coordinate square differences are equivalent to log-supermodularity on a Boolean cube, by summing those differences along coordinate paths. Iterating the one-coordinate elimination proves that the marginal p is log-supermodular.

For ordered $u \leq v$ and $s \leq t$, put

$$\Delta(u, v; s, t) = f(u, s) + f(v, t) - f(u, t) - f(v, s) \geq 0.$$

If $D = f(a, b) + f(a', b') - f(a, b') - f(a', b)$, then the exact identity

$$\begin{aligned} L - D &= \Delta(-, a'; -, b) + \Delta(a, +; b', +) \\ &\quad + \Delta(a', +; -, b') + \Delta(-, a; b, +) \end{aligned}$$

proves $D \leq L$. Interchanging b, b' proves $-D \leq L$. No order between a, a' or between b, b' is required. $\square$

The extreme rectangle has a direct Ising interpretation: restrict all spins of each terminal set to a common sign, thereby contracting that set to one vertex. The resulting two-spin interaction is unchanged by shifts of the terminal fields. Balancing those fields will let us apply the arbitrary-field covariance inequality of Ding–Song–Sun.

Lemma 2.2 (Pointwise ratios with arbitrary common pins). *Consider a finite ferromagnetic Ising graph, disjoint nonempty vertex sets I, J , and any fixed spins on a third set F . Delete F , retaining its contribution as fields, and contract I and J separately. Let q_0 be the zero-field FK probability that the two contracted vertices are connected on this contracted, vertex-deleted graph. Then every two conditional spin laws on I , obtained by prescribing arbitrary configurations b, b' on J , satisfy*

$$\exp(-4 \operatorname{arctanh} q_0) \leq \frac{\mu(\sigma_I = a \mid \sigma_J = b, \sigma_F)}{\mu(\sigma_I = a \mid \sigma_J = b', \sigma_F)} \leq \exp(4 \operatorname{arctanh} q_0) \quad \text{for every } a. \quad (2.2)$$

The same inequalities hold if either conditional law is replaced by any mixture of these laws. The assertion remains valid in the presence of additional arbitrary finite external fields; those fields are set to zero in the definition of q_0 .

Proof. A prescribed common spin τ_v , $v \in F$, contributes $K_{uv}\tau_v$ to the field at each surviving neighbor u . After these fields have been included, remove the vertices of F . Apply Lemma 2.1 to the joint marginal on I, J . Restricting both terminal sets to constant configurations is exactly their separate contraction. Its two-spin marginal has unnormalized weights

$$w(s, t) = \exp(C + Kst + As + Bt), \quad s, t \in \{-1, +1\}, \quad K = L/4 \geq 0.$$

The constants A, B include all effective fields from the other vertices. Add external fields $-A, -B$ at the two contracted vertices. This changes their marginal to weights proportional to e^{Kst} , without changing K . Both means now vanish and the covariance is $\tanh K$. The Ding–Song–Sun inequality applies to this finite ferromagnetic graph with these arbitrary mixed-sign fields [5, Corollary 1.3]. It gives

$$\tanh K \leq \langle \sigma_I \sigma_J \rangle_0 = q_0.$$

The last equality is the zero-field Edwards–Sokal identity [7]. Internal edges of a contracted terminal contribute constants and parallel edges have nonnegative summed coupling, so all hypotheses of that inequality are preserved. Thus $L \leq 4 \operatorname{arctanh} q_0$.

For fixed b, b' , Lemma 2.1 bounds the oscillation in a of the logarithm of the conditional density ratio by L . The ratio has mean one under its denominator law; its minimum is therefore at most one and its maximum at least one. The ratio lies throughout $[e^{-L}, e^L]$, as claimed. Averaging a pointwise inequality proves the mixture assertion, first for one law and then for the other. $\square$

Remark 2.3 (Symmetric terminals). The extreme log-odds rectangle and the connection probability q_0 are unchanged when I and J are interchanged. Thus the pointwise comparison applies with the observed and prescribed terminal sets reversed. Common pins are still deleted from the comparison graph; oppositely signed common pins are never wired together. This symmetry will also be used when a small region of changed pins is the source and the observed set is much larger.

2.2 Critical collars, small patches, and local charts

We now set every retained nearest-neighbor coupling to β_c . To bound the connection probability in Lemma 2.2, it suffices to find a dual circuit separating the terminals. A fixed collar provides a probability bounded away from zero; many disjoint collars give a power of the ratio of scales.

Common prescribed spins still contribute fields in the Ising law and are deleted only in the zero-field comparator.

We use the random-cluster representation with $p_c = 1 - e^{-2\beta_c}$ and cluster weight 2. An FK boundary wiring is a partition of the boundary vertices. An open crossing here always uses actual lattice edges; an abstract boundary identification is not itself a crossing. Free clusters receive independent fair spin signs in the Edwards–Sokal representation [7], and a cluster attached to plus pins has sign plus. We use the FK domain Markov property, association, and monotonicity under edge addition and boundary wiring in this finite-graph setting; these properties are recalled in [6, Section 2]. Their association and ordering principles originate in [8, 9]. Association applies to decreasing events too. Thus overlapping crossings used to construct a dual circuit can be combined by association, while disjoint tests can also be combined conditionally.

Lemma 2.4 (Conditional crossing tests). *Fix $A \geq 1$. There is $c_A > 0$ such that a lattice rectangle with both side lengths at least one and aspect ratio in $[A^{-1}, A]$ has probability at least c_A of either specified open crossing, and of either specified dual crossing, at criticality under any boundary wiring. Changing each side by a bounded number of lattice spacings only changes the constant.*

These lower bounds also apply to a decreasing dual-crossing test in a vertex-deleted graph, conditional on all bonds outside its full rectangular test region, provided no terminal identification has vertices in the interior of that region. Additional vertex deletions preserve the bound. For finitely many tests with disjoint sets of unexposed edges, the conditional bounds may be multiplied.

Proof. The rectangle assertion is the critical FK-Ising RSW theorem [6, arXiv version 1, Theorem 1], with planar duality for dual crossings. The domain Markov property turns an exterior bond configuration into a boundary wiring. Restore the deleted vertices of the test rectangle, restore their edges, and wire its entire outer boundary. The resulting measure stochastically dominates the conditional original bond law, so a decreasing dual event is at least as likely in the original law. The domination follows, for example, from the one-edge conditional probabilities p_c and $p_c/(p_c + 2(1 - p_c))$, according as the endpoints are or are not already connected: adding edges or wiring increases these probabilities. Restoring isolated vertices first and then their edges has the same effect. The hypothesis about terminals ensures that no identification in the interior of the restored rectangle is being discarded.

For the last assertion, order the test regions. Conditioning on all edges outside the next region includes the results of all earlier tests. Applying the uniform bound and then averaging proves the product lower bound by induction. The corresponding product upper bound for simultaneous failure of tests follows in the same way. If a bounded lattice correction makes a side have bounded length, finitely many prescribed closed or open edges, using the same one-edge probabilities, supplies a uniform replacement test. $\square$

Corollary 2.5 (Buffered and small-patch comparisons). *Fix $c > 0$. There is $K_c < \infty$ such that, on the square lattice, $I \subseteq z + B_R$ and $J \subseteq (z + B_{(1+c)R})^c$ imply pointwise density ratios in $[K_c^{-1}, K_c]$ for the conditional laws in Lemma 2.2. Other common pins may occur anywhere off $I \cup J$.*

There are absolute $K < \infty$ and $\delta > 0$ such that, if $I \subseteq z + B_w$, $J \subseteq (z + B_R)^c$, and $R \geq 8w \geq 8$, the logarithm of every such conditional density ratio has absolute value at most

$$K(w/R)^\delta. \tag{2.3}$$

In particular the total variation distance is at most $K'(w/R)^\delta$, and the expectation difference of a bounded test function is at most $2K'\|h\|_\infty(w/R)^\delta$. No free neighborhood of the source patch I is required.

The buffered comparison also holds between laws from different domains, including plane, torus, or cylinder laws, whenever their common collar is an isometric square-lattice chart and all differing data lie beyond it. It holds after applying the same conditional randomization kernel to local spins, in particular internal Edwards–Sokal bond sampling.

Proof. An annulus of fixed radius ratio separating the two terminals contains a dual circuit with probability at least $c_1(c) > 0$. Cover a middle subannulus by a fixed number of overlapping rectangles, require dual crossings in both directions of their overlap squares, and use Lemma 2.4. The crossings join to an actual dual circuit, which separates the terminals even after their separate internal wiring. Extra common pins are deleted in the FK comparator; restoring them only increases connectivity. Thus $q_0 \leq 1 - c_1(c)$, giving the first assertion by Lemma 2.2.

Between B_{2w} and $B_{R/2}$, choose annuli with radius ratio, say, four and disjoint test edges. There are at least $c_2 \log(R/w) - c_3$ such annuli. A connection must avoid the dual circuit in each, so Lemma 2.4 gives $q_0 \leq C_1(w/R)^{\delta_1}$. The fixed annulus between the terminals also gives $q_0 \leq 1 - c_4$. On $[0, 1 - c_4]$, $\operatorname{arctanh} q \leq C_2 q$. This proves the logarithmic bound. The total variation and expectation bounds follow by integrating the density difference.

For different domains, condition on the exterior neighbor spins of a square contained in the common chart. Each local law is a mixture of the same finite-volume conditional laws, so the mixture assertion applies. Finite-volume approximation gives the plane and cylinder statements; the bounds are uniform in the approximating volume. Finally, applying the same Markov kernel preserves pointwise domination of measures. $\square$

Corollary 2.6 (Local charts and finite patch covers). *The preceding buffered and patch bounds remain valid in a chart clipped by a free or a fixed physical boundary, provided the local graph and the common prescribed boundary spins agree in the compared laws. Deleted vertices and missing edges may occur in the chart. For laws on different domains, assume that conditioning on spins beyond the common chart gives the same local interaction and pinning rule.*

More generally, suppose the observed set is partitioned into at most m patches. If each patch has a square-lattice chart separating it from every changed datum by a fixed positive relative distance, the whole observed law has a pointwise density comparison with constant equal to the product of the patch constants. The same conclusion holds if instead the changed source set has such a partition, with each source patch separated from the whole observed set. Constants depend on m and the fixed chart clearances. Polygonal and rotated collars are therefore allowed whenever they admit such a finite cover.

Proof. First fix a common finite local conditional graph. In its zero-field FK comparator restore any vertices and edges missing from the full square test rectangles. Restoration increases connectivity. The decreasing dual tests in Lemma 2.4 therefore give the same upper bound on q_0 as before. Prescribed physical-boundary vertices are among the common pins: delete them in the comparator before this restoration. No identification is added in a test rectangle’s interior. Lemma 2.2 proves the asserted pointwise bounds for this conditional graph.

For different volumes or continuations, condition at the edge of the common chart. Both laws are mixtures of the same local kernels, with possibly different mixing measures. Apply the mixture assertion of Lemma 2.2. The restoration argument bounds the auxiliary connection probability; the common conditional kernel is what permits the comparison between volumes.

For a cover of the observed set, reveal its patches in a fixed order and factor each configuration probability by the chain rule. In the comparison for the next patch, previously revealed patches are common pins. They can lie anywhere off the two terminals. Each conditional ratio has its stated bound, and multiplication gives the product constant. For a cover of the source, telescope the

changes in its patches; all other source patches are common pins. Apply Remark 2.3 to place the small patch in the inner terminal. Mixtures over source assignments and common randomizations preserve the resulting bounds. $\square$

3 Spin correlations and magnetic moments

The magnetic estimates in this section concern the critical nearest-neighbor Ising model with no external fields except those induced by prescribed spins. Pins outside the stated free neighborhoods may have arbitrary signs.

Lemma 3.1 (Static correlation inputs). *The following statements hold with constants depending only on the indicated fixed bulk separations and aspect ratios.*

1. *In a square of radius r , under any FK boundary wiring,*

$$cr^{-1/8} \leq \phi(x \longleftrightarrow \partial B_r) \leq Cr^{-1/8}$$

for x in a fixed central subsquare. In the plus Ising law this is the corresponding magnetization estimate.

2. *In a plus rectangle of scale r and fixed aspect ratio, for two distinct points in separated fixed compact bulk sets,*

$$cr^{-1/4} \leq \text{Cov}_+(\sigma_x, \sigma_y) \leq Cr^{-1/4}.$$

3. *The infinite-volume zero-field law on $\mathbb{Z}^2$ is unique and spin-flip symmetric. For a lattice-dependent constant $c_\sigma > 0$,*

$$\mathbb{E}(\sigma_x \sigma_y) = c_\sigma |x - y|^{-1/4} (1 + o(1)) \quad (|x - y| \rightarrow \infty), \quad 0 \leq \mathbb{E}(\sigma_x \sigma_y) \leq C(1 + |x - y|)^{-1/4}.$$

The asymptotic is uniform in direction.

Proof. The centered square assertion is [6, arXiv version 1, Lemma 26]. For another bulk point, take a square of radius $c'r$ about that point contained in the original square. Its free-boundary arm lower bound remains a lower bound for the arm inside the larger domain. Require in addition an open circuit in its outer half-annulus and a rectangle crossing from its inner half to the original boundary. The arm, circuit, and crossing then join. RSW gives the latter two events jointly a fixed positive probability, and FK association multiplies this by the arm lower bound. The upper bound uses the wired arm in a contained square. This proves the stated uniformity. The magnetization tends to zero as the boundary recedes. The monotone coupling of plus and minus boundary laws has disagreement probability equal to half their mean difference at a vertex. A union bound on any fixed finite set therefore makes the extremal infinite-volume laws equal, proving uniqueness and symmetry.

For the second assertion, apply bulk one- and two-spin convergence [4, Theorem 1.3] in the fixed rescaled rectangle. We spell out strict positivity of the limiting connected function. In the upper half-plane let $q = |z - z'|/|z - \bar{z}'| \in (0, 1)$. The plus one- and two-spin formulae [4, Theorem 7.1, Equation (7.3)] give

$$\frac{\langle \sigma_z \sigma_{z'} \rangle_+}{\langle \sigma_z \rangle_+ \langle \sigma_{z'} \rangle_+} = \left(\frac{q^{1/2} + q^{-1/2}}{2} \right)^{1/2} > 1.$$

Conformal covariance transfers this strict inequality to a rectangle. Continuity and compactness give a positive minimum for the connected function on separated compact bulk sets. Uniform bulk

convergence gives the discrete lower bound for sufficiently large r ; finite energy absorbs the finitely many smaller scales. The upper bound also follows from convergence, or from two disjoint wired arm tests.

For completeness, put $R = |x - y|$ and $u = (y - x)/R$, and bracket the plane two-point function between free and plus laws in a square centered at x of radius TR . Fix $T > 2$ first and take $R \rightarrow \infty$. Theorem 1.3 of [4] gives the continuum limits $F_T^b(0, u)$, for $b \in \{\text{free}, +\}$, uniformly for $|u| = 1$. Conformal scaling gives $F_T^b(0, u) = T^{-1/4} F_1^b(0, u/T)$. The normalization and two-spin fusion in [4, Proposition 5.3, Theorem 6.3, Equation (6.12)] imply $F_1^b(0, u/T) = T^{1/4}(1 + o_T(1))$ for both boundary conditions. Remark 6.4 there makes the error uniform in direction: the colliding pair remains in the bulk and there are no other insertions. Letting $T \rightarrow \infty$ after the fixed- T mesh limit proves the plane asymptotic with the common lattice conversion constant c_σ , uniformly in direction. Finally, connection of x to y forces arms in the two disjoint boxes of radius $\lfloor |x - y|/4 \rfloor$. Conditioning on all other edges and using the wired upper arm bound proves the uniform upper bound. Nonnegativity follows from the FK representation. $\square$

Lemma 3.2 (Product arm bound). *Let Q be a lattice square with an ordinary FK bond law under arbitrary boundary wiring. After sampling the bonds, assign each cluster touching the boundary any sign, and give every other cluster an independent fair sign. Conditional on the bonds, all nonboundary cluster signs are mutually independent and independent of the entire collection of boundary-cluster signs. Boundary-cluster signs may have any joint law conditional on the bonds; the bonds are not further conditioned on compatibility with prescribed mixed signs. Suppose $x_1, \dots, x_n$ lie a distance at least cs from ∂Q and in a set of diameter at most As , where $n \geq 2$. For*

$$r_i = \max \left(1, \left\lfloor \min \left(\frac{cs}{4}, \frac{1}{4} \min_{j \neq i} \|x_i - x_j\|_\infty \right) \right\rfloor \right),$$

one has

$$\left| \mathbb{E} \prod_{i=1}^n \sigma_{x_i} \right| \leq C_{n,c} \prod_{i=1}^n r_i^{-1/8}.$$

The same bound holds conditionally on arbitrary bonds exterior to Q , with the induced wiring, before cluster colors are sampled.

Proof. First cancel insertions that occur an even number of times at one vertex. At a vertex occurring repeatedly, the original r_i 's are one, so discarding those factors costs nothing. All remaining radii are the original radii; none is recomputed after cancellation. Each remaining singleton whose ball has radius larger than one must have an open arm to that ball's boundary if the conditional cluster-sign average is nonzero: otherwise its free cluster contains no other odd insertion and has mean sign zero. A cluster reaching a prescribed-sign boundary also requires this arm. The balls of radius greater than one are disjoint, and are contained in Q . Their arm probabilities, conditionally on all exterior edges, are at most their fully wired probabilities. Lemma 3.1 bounds each by $Cr_i^{-1/8}$. Sequential conditioning multiplies the bounds. Absolute cluster-sign averages are at most one. Radius-one balls and parity cancellations can only remove requirements, proving the claim. Exterior bond conditioning supplies only boundary wiring, so the same proof applies conditionally. Prescribed mixed-spin boundaries are handled by the comparison in Lemma 3.3. $\square$

Lemma 3.3 (Buffered line and area moments). *Fix $A, c > 0$, an integer $p \geq 1$, and $\nu \in \{1, 2\}$. Let S be a set of lattice points of diameter at most As such that*

$$|S \cap B_t(x)| \leq A(t+1)^\nu \quad (t \geq 0), \quad |S| \leq As^\nu.$$

Assume the Ising law has a free square-lattice neighborhood of width cs about S , with arbitrary pins beyond that neighborhood. Then, for weights $|a_x| \leq 1$,

$$\mathbb{E} \left| \sum_{x \in S} a_x \sigma_x \right|^{2p} \leq C_{p,A,c} s^{2p(\nu-1/8)}. \quad (3.1)$$

This includes a fixed number of line segments or square contours ($\nu = 1$), and area sums ($\nu = 2$). It applies as well in a buffered chart of a torus or cylinder. No pin on S is allowed in this assertion.

Proof. Use Corollary 2.5, and a fixed finite covering if necessary, to compare to a plus square with a proportional buffer. It is enough to prove the bound there. Expand the even power and take absolute values of the spin-product expectations. Lemma 3.2 and the diameter bound reduce the desired sum to a constant times

$$\sum_{x_1, \dots, x_{2p} \in S} \prod_{i=1}^{2p} \sum_{j \neq i} (1 + \|x_i - x_j\|_\infty)^{-1/8}.$$

Indeed the capped nearest distance in that lemma is bounded below by a fixed multiple of the uncapped nearest distance plus one, since all points lie within diameter As .

Expand the product by choosing one neighbor $j(i) \neq i$ for each index. The directed graph on the indices has outdegree one, and each component consists of an oriented cycle of length at least two with in-trees attached. Put $q = 1/8$. The counting hypothesis gives, uniformly in y ,

$$\sum_{x \in S} (1 + \|x - y\|_\infty)^{-q} \leq Cs^{\nu-q}, \quad \sum_{x \in S} (1 + \|x - y\|_\infty)^{-2q} \leq Cs^{\nu-2q}.$$

These follow by dyadic distance shells; both q and $2q$ are less than ν . Summing an in-tree leaf removes one edge and contributes $Cs^{\nu-q}$. Once only a cycle remains, sum one vertex incident to two cycle edges by Cauchy–Schwarz, at cost $Cs^{\nu-2q}$. For a two-cycle the two edges have the same endpoints and this is the second displayed bound. The remaining graph is a path; remove its leaves at cost $Cs^{\nu-q}$ each, and its last vertex at cost Cs^ν . A component with m vertices therefore costs at most $C_m s^{m(\nu-q)}$. There are at most $(2p-1)^{2p}$ neighbor choices, proving the moment estimate.

For a large or bent support, partition S by overlay squares of side $cs/16$. At most $C(1+A/c)^2$ such squares meet S . In each nonempty piece choose $z \in S$ in that square. The piece lies in $z + B_{cs/8}$, and $z + B_{cs/2}$ lies in the assumed free neighborhood because $z \in S$. Compare only this local marginal to the plus law in the latter square. The counting hypotheses persist with constants depending only on A, c . Apply the proved bound to each piece and use the triangle inequality in L^{2p} . Lattice rounding at bounded cs is absorbed in the same constants. The same construction works in periodic charts. $\square$

4 Ordered transmission across a cut

The pointwise comparison controls the effect of changing one small patch of a prescribed boundary. We now turn this into a bound for an ordered change along an entire separating cut. The quantity on the right will be the change of its normalized magnetization. This lets a one-point estimate control the law of every bounded observable beyond the cut.

4.1 The cut geometry and the transmission estimate

For the lattice statements, each local chart carries the critical nearest-neighbor Ising model. No external fields are added beyond those induced by prescribed spins, whose common signs remain arbitrary. We state the geometry in a form which records all uniformity requirements. A source S and target T are disjoint vertex sets. Common pins away from them are allowed. Delete the common pinned vertices from the graph, retaining S as boundary vertices. A finite set J , disjoint from the source, target, and common pins, is a *separating cut* if every path from S to T in this graph visits J . The continuation on the target side of J is then independent of the prescribed spins on S .

We will partition J into small patches b_i . The pointwise comparison bounds the effect of changing one patch; a straight counting segment L_i in the same neighborhood will record the corresponding magnetic change. The next definition provides free space for this local count and limits how often one spin can be counted.

Definition 4.1 (An admissible cut at scales w, s). Fix constants $A, K \geq 1$ and $a \in (0, 1/(16A))$. For $0 < w \leq s$, a separating cut J is admissible if it has a partition into patches (b_i) and associated straight counting segments $L_i \subseteq J$, with centers z_i and scales $\ell_i \in [aw, Aw]$, satisfying:

1. $b_i \cup L_i \subseteq z_i + B_{\ell_i}$ and $|L_i| \in [a\ell_i, 2\ell_i]$;
2. every $z_i + B_{16\ell_i}$ is a pin-free planar lattice chart, including freedom from source pins;
3. each site of J belongs to at most K counting segments;
4. T is at distance at least s/A from every z_i , $16\ell_i \leq s/A$, and the patch comparison of Corollary 2.5 is applicable in a chart up to a fixed positive multiple of s at every patch.

All constants in the following statements may depend on a, A, K and the fixed chart geometry, but not on w, s , the number of source pins, their values, or the mesh. Decreasing a or subdividing the patches by a fixed factor preserves this convention.

The principal examples are concentric square contours in a common free collar of width comparable to w , separated from the target by a fixed multiple of s , and parallel periodic rows with the same properties. For a square, partition each side into intervals of length a sufficiently small fixed multiple of w . Near a corner, choose a counting segment on one of its adjacent sides. Taking these intervals smaller than the collar width by a fixed factor gives the free squares in Definition 4.1; a bounded number of neighboring intervals can use the same segment, giving a uniform overlap bound. Fixed finite unions of these cuts are allowed. The hypotheses are also appropriate when w/s is small: the constants just described do not change with that ratio.

Write

$$M_J = s^{-7/8} \sum_{x \in J} \sigma_x. \quad (4.1)$$

The normalization always uses the displayed reference scale s , not the individual patch scale.

Proposition 4.2 (Transmission). *There is an exponent $\delta \in (0, 1/8)$, chosen by decreasing the exponent of Corollary 2.5 if necessary, with the following property. Prescribe two ordered configurations on S , use identical common pins elsewhere, and let Δ denote higher-law expectation minus lower-law expectation. For an admissible separating cut at scales w, s , a bounded observable h supported on T , and w sufficiently large,*

$$|\Delta h| \leq C \|h\|_{\infty} (w/s)^{-7/8+\delta} \Delta M_J. \quad (4.2)$$

Equivalently, if $\|h\|_\infty \leq H$ for fixed $H > 0$,

$$\Delta M_J \geq c_H (w/s)^{7/8-\delta} |\Delta h|. \quad (4.3)$$

The constants are uniform as w/s decreases within the stated geometry.

The exponent in (4.2) comes from two estimates. A changed patch of diameter comparable to w changes the conditional expectation of h by at most $C\|h\|_\infty(w/s)^\delta$, by the small-patch comparison. We construct an ordered coupling in which, conditional on a disagreement reaching that patch, the expected number of disagreements on its counting segment is at least $cw^{7/8}$. Bounded overlap sums these contributions over J . The normalization $s^{-7/8}$ then gives the factor $(w/s)^{-7/8+\delta}$.

We prove the second estimate first. An auxiliary FK representation will identify ordered disagreements with connections to the changed pins. A local domination statement will let us test for magnetic mass inside the pin-free charts of Definition 4.1. We will then return to the cut and prove the displayed transmission estimate.

4.2 An auxiliary FK model for ordered disagreements

Begin on a finite graph $\mathcal{G} = (V, E)$ with couplings $K_e \geq 0$ and a pin set $P \subseteq V$. For a retained free set $H \subseteq V \setminus P$, at specified couplings J_e let $Z_{P,\tau}(H)$ denote the Ising partition sum on the induced graph $P \cup H$, with signs τ on P . Partition sums include edges between pinned vertices. Their contribution is constant in the free spins. In the planar applications the original couplings are $K_e = \beta_c$. The FK bonds introduced below instead have doubled couplings on a random induced set; the critical FK crossing theorem is not applied directly to these auxiliary bonds.

Take two independent Ising configurations η, ξ whose signs on the same pin set P are ordered, with the higher signs assigned to η . Let $U \subseteq P$ be the disagreement pins and $A = P \setminus U$ the agreement pins, whose common signs are denoted by τ . Let $W = V \setminus P$, and let $G \subseteq W$ be the set of free sites where the original replicas disagree. Thus the full auxiliary disagreement set is $G \cup U$.

Lemma 4.3 (Doubled representation and ordered coupling). *With partition sums at couplings $J_e = 2K_e$, the law of G is*

$$\mu_\tau(G) \propto Z_{U,+}(G) Z_{A,\tau}(W \setminus G). \quad (4.4)$$

Conditional on G , the spins $\eta = -\xi$ on $G \cup U$ form the first Ising law in (4.4), independently of the common spins on its complement. Put FK bonds ω on $G \cup U$. There is a coupling of the original two Ising marginals, denoted again by (η, ξ) , for which $\eta \geq \xi$ and

$$\eta_x - \xi_x = 2\mathbf{1}_{\{x \longleftrightarrow U\}}, \quad \mathbb{E}\eta_x - \mathbb{E}\xi_x = 2\mathbb{P}(x \longleftrightarrow U). \quad (4.5)$$

Here and below a connection to U uses actual open edges, and is false for $x \notin G \cup U$.

Proof. For an edge with endpoints in the same one of $G \cup U$ and its complement, the two replica energies add to twice the energy of their common orientation. For an edge between these sets they cancel. This proves (4.4) and the conditional independence. Edges with both endpoints pinned contribute constants independent of G , so cause no exception to the factorization.

Every FK cluster in $G \cup U$ avoiding U has an independent fair orientation. On each such cluster replace the pair of opposite signs by two copies of the first replica's sign. The first replica is unchanged; the second replica's collection of independent fair cluster signs has been negated, so its conditional marginal is unchanged as well. Clusters meeting U retain the pair $(+, -)$, and agreement sites retain equal signs. This proves both the ordering and (4.5). The auxiliary set G need not equal the disagreement set after this operation: the latter is precisely the union of clusters meeting U . $\square$

Order pairs (G, ω) by inclusion in both coordinates, padding the bond configuration by closed edges off $G \cup U$. An event or function of this pair is called increasing in this order. The *plus-agreement reference* is the law in which all signs on A have been changed to plus, while the disagreement pins U are retained.

The coupled replicas now differ exactly on clusters meeting U . The auxiliary set G still records the original independent replicas, and can contain clusters on which the coupled replicas agree. To obtain lower bounds on its local connections, we compare its law with the plus-agreement reference. The density between the two auxiliary set laws is, up to normalization,

$$r(G) = \frac{Z_{A,\tau}(W \setminus G)}{Z_{A,+}(W \setminus G)}.$$

We will show that this density increases when G grows. The needed partition-sum fact is a comparison of the factors gained by adding one free vertex. We prove it for arbitrary nonnegative couplings J_e , then apply it with $J_e = 2K_e$.

Lemma 4.4 (Spin products and addition of vertices). *Fix a set P of pinned vertices, couplings $J_e \geq 0$, and let $Z_{P,\tau}(H)$ be the Ising partition sum on the induced graph with vertex set $P \cup H$, with signs τ on P . Here $H \subseteq V \setminus P$ is the retained free set. Then the following assertions hold.*

1. *With all pins plus, every spin-product expectation is nonnegative and increases when nonnegative couplings are increased, free vertices and their incident ferromagnetic edges are added, or additional spins are pinned plus. The product in question involves only vertices present before the change.*
2. *For every finite set $Q \subseteq P \cup H$,*

$$\left| \mathbb{E}_{P,\tau,H} \prod_{x \in Q} \sigma_x \right| \leq \mathbb{E}_{P,+,H} \prod_{x \in Q} \sigma_x. \quad (4.6)$$

3. *The function $H \mapsto Z_{P,+}(H)$ is log-supermodular. Moreover, for $v \notin P \cup H$, the site gains satisfy*

$$\frac{Z_{P,\tau}(H \cup \{v\})}{Z_{P,\tau}(H)} \leq \frac{Z_{P,+}(H \cup \{v\})}{Z_{P,+}(H)}. \quad (4.7)$$

Proof. Use the FK representation with edge parameters $p_e = 1 - e^{-2J_e}$. In descriptions of sign compatibility, an *actual cluster* means a component formed by open edges of the original graph: virtual wires between pins do not count as open paths. Write $k_0(\omega)$ for the number of actual clusters avoiding P . Apart from a constant, the all-plus bond weight is

$$\prod_{e \in E(P \cup H)} p_e^{\omega_e} (1 - p_e)^{1 - \omega_e} 2^{k_0(\omega)}. \quad (4.8)$$

This is the usual FK law with all pins wired. Conditional on the bonds, clusters touching a pin have sign plus, and the other clusters have independent fair signs.

Let $\mathcal{F}_Q$ be the event that each actual cluster avoiding P contains an even number of vertices of Q . Conditional sign averaging gives

$$\mathbb{E}_{P,+,H} \prod_{x \in Q} \sigma_x = \phi_{P,+,H}(\mathcal{F}_Q). \quad (4.9)$$

The event $\mathcal{F}_Q$ is increasing. Indeed, merging two unpinned clusters whose parities are even preserves their even parity, whereas merging any cluster with a pinned cluster removes its parity restriction. Adding an initially isolated vertex outside Q adds a cluster of even parity zero. Thus FK monotonicity under edge addition, increases of edge parameters, and increases of the boundary wiring proves the first assertion. Pinning an additional spin plus is covered by the same argument: its cluster ceases to have a parity restriction and the FK wiring increases. Positive fields, when needed, may equivalently be represented by edges to one additional plus vertex.

For mixed signs, put

$$\mathcal{D}_\tau = \{\text{no actual cluster contains two pins with opposite signs}\}.$$

The mixed-sign bond weight is exactly (4.8) multiplied by $\mathbf{1}_{\mathcal{D}_\tau}$. In particular, its law is $\phi_{P,+,H}(\cdot \mid \mathcal{D}_\tau)$. This identity also handles edges between opposite pins: such an edge must be closed. The event $\mathcal{D}_\tau$ is decreasing and has positive probability. Conditional sign averaging in the mixed law has absolute value at most $\mathbf{1}_{\mathcal{F}_Q}$. Positive association of the wired FK law therefore gives

$$\left| \mathbb{E}_{P,\tau,H} \prod_{x \in Q} \sigma_x \right| \leq \phi_{P,+,H}(\mathcal{F}_Q \mid \mathcal{D}_\tau) \leq \phi_{P,+,H}(\mathcal{F}_Q),$$

which proves (4.6).

For the last assertion, summing the spin at a newly added vertex gives the exact identity

$$g_{P,\tau,H}(v) := \frac{Z_{P,\tau}(H \cup \{v\})}{Z_{P,\tau}(H)} = 2\mathbb{E}_{P,\tau,H} \cosh \left(\sum_{u \in P \cup H} J_{vu} \sigma_u \right). \quad (4.10)$$

Zero couplings may be omitted from the sum. If N is its neighbor set, the integrand has the expansion

$$\cosh \left(\sum_{u \in N} J_{vu} \sigma_u \right) = \prod_{u \in N} \cosh J_{vu} \sum_{\substack{Q \subseteq N \\ |Q| \text{ even}}} \prod_{u \in Q} \tanh J_{vu} \prod_{u \in Q} \sigma_u. \quad (4.11)$$

All coefficients are nonnegative. Under plus pins, enlarging H increases the old product expectations, adds nonnegative terms, and increases the prefactor. Hence $g_{P,+,H}(v)$ increases with H . Increasing site gains are equivalent to log-supermodularity: add the vertices of $H_1 \setminus H_2$ one at a time, comparing their gains above $H_1 \cap H_2$ and above H_2 , to obtain

$$Z_{P,+}(H_1 \cup H_2) Z_{P,+}(H_1 \cap H_2) \geq Z_{P,+}(H_1) Z_{P,+}(H_2).$$

Finally, apply (4.6) term by term in (4.11) to prove (4.7). $\square$

Lemma 4.5 (Association and increasing conditioning). *The plus-agreement reference law of (G, ω) is positively associated. The mixed-agreement law has an increasing density, depending only on G , relative to this reference. Consequently, if $\mathcal{E}$ is any increasing event of positive probability and f is any increasing function, then*

$$\mathbb{E}_\tau[f \mid \mathcal{E}] \geq \mathbb{E}_+[f]. \quad (4.12)$$

There is no restriction on the number or placement of the pins in this finite-graph assertion.

Proof. By Lemma 4.4, both factors in (4.4) with plus pins are log-supermodular functions of G . Complementation exchanges unions and intersections and hence preserves this property. Their

product is log-supermodular, so the reference set law is positively associated. Conditional on G , the FK law is positively associated and increases stochastically with G . Indeed, one may first add isolated vertices and then increase their edge parameters from zero. Thus conditional expectations of increasing functions of (G, ω) increase with G . The decomposition of a covariance into its conditional covariance and the covariance of its conditional expectations proves joint positive association.

The mixed law has the density $r(G)$ displayed above. By (4.7), the quotient $Z_{A,\tau}(H)/Z_{A,+}(H)$ decreases as H grows. Therefore r increases with G . The conditional law on the left side of (4.12) has density proportional to $r\mathbf{1}_{\mathcal{E}}$ against the associated reference. That density is increasing, and association gives

$$\frac{\mathbb{E}_+[fr\mathbf{1}_{\mathcal{E}}]}{\mathbb{E}_+[r\mathbf{1}_{\mathcal{E}}]} \geq \mathbb{E}_+[f].$$

This argument requires neither association of the mixed law nor preservation of a lattice condition after increasing-event conditioning. $\square$

Proposition 4.6 (Local domination). *Let $Q \subseteq W$ contain no pins. In Q , form the doubled model from two independent Ising configurations with plus signs on every exterior neighbor of Q , with the original couplings K_e . Call its auxiliary set-and-bond law $\mathbb{P}_Q^{++}$. For every increasing function f of the auxiliary set and bonds internal to Q , and every positive-probability increasing event $\mathcal{E}$ in the original doubled model,*

$$\mathbb{E}_{\tau}[f \mid \mathcal{E}] \geq \mathbb{E}_Q^{++}[f]. \quad (4.13)$$

The same statement holds with $\mathcal{E}$ the sure event.

Proof. By Lemma 4.5, it suffices to compare the plus-agreement reference with $\mathbb{P}_Q^{++}$. Condition its exterior auxiliary set $G \setminus Q$. Fix $v \in Q$ and the membership of all other sites of Q . Write D for the full disagreement vertices excluding v , including the pins U , and H for the full agreement vertices excluding v , including the pins A . The conditional odds for membership of v are the quotient of two site gains:

$$\frac{\mu_+(v \in G \mid G \setminus \{v\})}{\mu_+(v \notin G \mid G \setminus \{v\})} = \frac{g_{U,+,D \setminus U}(v)}{g_{A,+,H \setminus A}(v)}. \quad (4.14)$$

Here g is the gain in (4.10).

The numerator is at least the gain on the induced disagreement set in Q : adding the exterior disagreement vertices, their edges, and their plus pins only increases it. For the denominator, first pin every exterior agreement vertex plus. The parity argument in Lemma 4.4 and the nonnegative expansion (4.11) show that this increases the gain. Then restore every missing exterior neighbor of Q as a plus pin, again increasing the gain. The resulting denominator is exactly the agreement-site gain for the box-alone model: partition factors involving only the now-pinned exterior cancel in the site ratio. Thus the odds in (4.14) are at least the corresponding odds of $\mathbb{P}_Q^{++}$ at every fixed interior set configuration.

Equivalently, the density of the reference interior-set law relative to the box-alone set law is increasing: the ratio increases under every single-site addition. The box-alone set law is associated by Lemma 4.5. Tilting it by this increasing density therefore proves stochastic domination of the sets.

Given a full auxiliary set, conditioning the exterior FK bonds induces some wiring on its part in Q . This wiring dominates the free boundary condition of the box-alone disagreement-bond system; its orientation has no disagreement pins in Q or on its exterior. The free FK law on an induced set also grows with that set. First perform this bond comparison and then the set comparison just obtained. Averaging over the exterior set proves the claim for every increasing f . $\square$

4.3 Mass carried by a disagreement connection

Local domination remains valid after conditioning on a connection from the disagreement pins. We now construct, in the doubled plus-boundary reference, an increasing event joining many sites of a segment to such a connection. This is where the correlation and moment estimates of Section 3 enter.

For the remainder of the section the underlying graph in each local chart is the square lattice at β_c . Bounded integer roundings of rectangles and segments are understood. Every lower bound below holds once its scale is sufficiently large; the threshold depends only on the fixed geometric constants. This is sufficient when macroscopic geometry is fixed before taking the mesh limit.

The bonds in this subsection belong to the auxiliary set sampled before the recoloring in Lemma 4.3. Equal plus boundary assignments can therefore yield auxiliary bond connections, even though the resulting ordered spin coupling has identical replicas.

Lemma 4.7 (Many sites joined across a rectangle). *Consider two independent plus-boundary Ising configurations in a rectangle of scale w and fixed aspect ratio, and their auxiliary disagreement bonds. Let L_1, L_2 be straight lattice segments of lengths comparable to w , in fixed compact bulk subsets of the rectangle, mutually separated by a distance comparable to w . There are constants $c, p > 0$, depending only on this geometry, such that*

$$\mathbb{P}_Q^{++}(|\{x \in L_1 : x \longleftrightarrow L_2\}| \geq cw^{7/8}) \geq p. \quad (4.15)$$

In particular a disagreement-bond connection between the segments has probability bounded below.

Proof. In the original independent pair put $Y_x = (\eta_x - \xi_x)/2$. There are no disagreement pins. Conditional cluster signs give

$$\mathbb{P}_Q^{++}(x \longleftrightarrow y) = \mathbb{E}[Y_x Y_y] = \frac{1}{2} \text{Cov}_Q^+(\sigma_x, \sigma_y). \quad (4.16)$$

More generally,

$$\mathbb{P}_Q^{++}(x \longleftrightarrow y, x' \longleftrightarrow y') \leq \mathbb{E}[Y_x Y_y Y_{x'} Y_{y'}]. \quad (4.17)$$

Indeed, if both pairs connect, all auxiliary clusters contain an even number of these four indexed factors. Conditional sign averaging of the product is the indicator of this even-parity event, with the additional requirement that all factors lie in the auxiliary set. This explanation also proves (4.17) when indices repeat.

Set

$$X = \sum_{x \in L_1} \sum_{y \in L_2} \mathbf{1}_{\{x \longleftrightarrow y\}}.$$

The bulk connected-correlation lower bound in Lemma 3.1 gives $\mathbb{E}X \geq c_1 w^{7/4}$. By (4.17), Cauchy–Schwarz, and the fourth line moments in Lemma 3.3,

$$\mathbb{E}X^2 \leq \mathbb{E} \left[\left(\sum_{L_1} Y_x \right)^2 \left(\sum_{L_2} Y_y \right)^2 \right] \leq C_1 w^{7/2}.$$

The moments for Y follow from those for either replica by the triangle inequality in L^4 . Paley–Zygmund therefore gives constants $c_2, p_2 > 0$ such that $\mathbb{P}(X \geq c_2 w^{7/4}) \geq p_2$.

Let L be the largest number of sites of L_2 in one auxiliary cluster. Then

$$L^2 \leq \sum_{\mathcal{C}} |\mathcal{C} \cap L_2|^2, \quad \mathbb{E} \sum_{\mathcal{C}} |\mathcal{C} \cap L_2|^2 = \mathbb{E} \left(\sum_{L_2} Y_x \right)^2 \leq C_2 w^{7/4}.$$

Choose K so large that $C_2/K^2 < p_2/2$. With probability at least $p_2/2$, both $X \geq c_2 w^{7/4}$ and $L \leq K w^{7/8}$ hold. Every site of L_1 reaching L_2 contributes at most L to X . On that intersection at least $(c_2/K)w^{7/8}$ sites reach L_2 , proving (4.15). Although the event involving L was used to establish its probability, the event in (4.15) itself is increasing. $\square$

Lemma 4.8 (A surrounding disagreement circuit). *In the doubled plus-boundary model on B_{16w} , a circuit of actual disagreement bonds in $B_{4w} \setminus B_{3w/2}$ surrounding $B_{3w/2}$ has probability at least a constant $p_0 > 0$ for all sufficiently large w .*

Proof. Consider the four strips

$$[-4w, 4w] \times [2w, 4w], \quad [-4w, 4w] \times [-4w, -2w], \quad [2w, 4w] \times [-4w, 4w], \quad [-4w, -2w] \times [-4w, 4w].$$

For the top strip, extend the long coordinate to $[-5w, 5w]$ and put agreeing plus boundary conditions on this extended rectangle. Apply Lemma 4.7 to vertical segments at $x = -4w$ and $x = 4w$, each with its y coordinate in $[5w/2, 7w/2]$. The segments are in its bulk. Their connection stays inside the extended rectangle, since the auxiliary disagreement set cannot contain its agreeing boundary pins. A subpath of this connection gives a left-to-right crossing of the unextended top strip. Proposition 4.6 transfers its positive lower bound to the doubled model on B_{16w} . The same construction applies to the other three strips.

The four crossing events are increasing, so their intersection has probability at least the product of their four lower bounds. On this intersection, adjacent crossings intersect inside their corner overlap: one crosses that overlap in one coordinate and the other in the perpendicular coordinate. The union contains a circuit surrounding the central square. One way to see the last assertion is to join successive intersection points along the four crossings, erase portions lying beyond the adjacent corner overlaps, and obtain a closed walk in the annulus with winding number one about the origin. Removing closed subwalks decomposes it into simple circuits; at least one retains nonzero winding number. All its edges lie in the four strips. This proves the claim. $\square$

Proposition 4.9 (Mass at a source-connected patch). *Fix $a > 0$. Let $b_0 \subseteq z + B_w$, and let $L \subseteq z + B_w$ be a horizontal or vertical lattice segment of length at least aw and at most $2w$. Suppose $z + B_{16w}$ is a free square-lattice neighborhood: it contains no common or disagreement pins. Then in the doubled model with arbitrary ordered pins outside this square,*

$$\sum_{x \in L} \mathbb{P}(x \longleftrightarrow U) \geq c_a w^{7/8} \mathbb{P}(U \longleftrightarrow b_0). \quad (4.18)$$

The constant and lower threshold for w depend only on a . The same statement holds inside an isometric planar chart of a torus or cylinder.

Proof. Translate the picture so that $z = 0$. Choose a translate L' of L by $6w$ in a perpendicular coordinate. Both segments lie in fixed bulk subsets of B_{16w} , while L' lies outside B_{4w} . Under its doubled plus-boundary law, Lemma 4.7 gives positive probability that at least $c w^{7/8}$ sites of L reach L' by paths using bonds internal to B_{16w} . Intersect this increasing local event with the increasing circuit event of Lemma 4.8. Association makes the probability of their intersection at least a positive constant p .

If $\mathbb{P}(U \longleftrightarrow b_0) = 0$, the claim is immediate. Otherwise condition on that increasing event and use Proposition 4.6. The circuit and the many-sites event still occur jointly with conditional probability at least p . Every path from U , which is outside B_{16w} , to $b_0 \subseteq B_w$ meets the surrounding circuit. Every counted path from $L \subseteq B_w$ to L' also meets it. By planarity all the counted sites are therefore connected to U . The conditional expected number of such sites is at least $p c w^{7/8}$. Multiplying by the probability of the conditioning event proves (4.18). $\square$

4.4 From local mass to transmission

Proposition 4.9 provides the local estimate promised at the start of the section. At each patch of an admissible cut, its scale $\ell_i \asymp w$ converts the probability of reaching that patch into expected disagreement mass on L_i . We now sum these estimates and use the common target-side continuation to complete the proof.

Proof of Proposition 4.2. Use the monotone coupling of Lemma 4.3. Let $\eta_J \geq \xi_J$ be its cut configurations. Put $k(j) = \mathbb{E}[h \mid \sigma_J = j]$, where the conditional expectation uses the common target-side continuation. The spatial Markov property and the separating property imply that this function does not depend on which source configuration was prescribed. Consequently

$$\Delta h = \mathbb{E}[k(\eta_J) - k(\xi_J)].$$

In this telescope, the other cut spins are common pins for the pointwise comparison. The local mass estimate is applied to the original coupling, whose counting charts remain pin-free.

Interpolate between the two cut configurations one patch at a time, using any deterministic ordering of their partition. Intermediate configurations are legitimate cut configurations; spins at all other patches are fixed for each comparison. Corollary 2.5 bounds the oscillation of k in one patch by $C\|h\|_\infty(w/s)^\delta$. It applies with these arbitrary other cut spins as common pins. If the two sampled cut configurations agree on the patch, the corresponding summand is zero. By (4.5), disagreement occurs somewhere in b_i exactly when $U \longleftrightarrow b_i$. Therefore

$$|\Delta h| \leq C\|h\|_\infty(w/s)^\delta \sum_i \mathbb{P}(U \longleftrightarrow b_i). \quad (4.19)$$

Proposition 4.9, applied with scale $\ell_i \asymp w$, and the overlap bound on the counting segments give

$$\sum_i \mathbb{P}(U \longleftrightarrow b_i) \leq Cw^{-7/8} \sum_i \sum_{x \in L_i} \mathbb{P}(U \longleftrightarrow x) \leq CKw^{-7/8} \sum_{x \in J} \mathbb{P}(U \longleftrightarrow x).$$

The last sum is $\frac{1}{2}s^{7/8}\Delta M_J$ by (4.5). Substitute it in (4.19) to obtain (4.2). All local constructions use the same fixed aspect ratios and overlap constants, independently of w/s . $\square$

Definition 4.10 (Order-positive signed source). For a probability law μ on the source spins, a real function q is an order-positive centered density if $\mathbb{E}_\mu q = 0$ and $\mathbb{E}_\mu[qf] \geq 0$ for every increasing source function f . The word density here refers to a signed measure; q need not be nonnegative or itself increasing.

Corollary 4.11 (Random and signed sources). *The conclusions of Proposition 4.2 hold after averaging over any ordered coupling of two random source configurations, with the appropriate common Gibbs continuation given each source. In particular, let μ be a Gibbs law with common pins and let $q = q(\sigma_S)$ be an order-positive centered density for its source marginal. Then*

$$|\mathbb{E}_\mu[qh]| \leq C\|h\|_\infty(w/s)^{-7/8+\delta} \mathbb{E}_\mu[qM_J]. \quad (4.20)$$

The constants do not depend on the mass of the positive part of q .

Proof. For an ordered random pair of sources, apply (4.2) to each pair and integrate. The magnetic differences are nonnegative. The triangle inequality bounds the absolute value of the integrated target difference by the integral of its absolute value, proving the first assertion.

For the signed assertion put $m = \mathbb{E}_\mu q_+ = \mathbb{E}_\mu q_-$. If $m = 0$ there is nothing to prove. Otherwise let $\mu_\pm = m^{-1} q_\pm \mu_S$ be probability laws on source configurations. Order positivity says $\mu_+(F) \geq \mu_-(F)$ for every increasing set F . These laws admit a coupling (s_+, s_-) with $s_+ \geq s_-$. For completeness, this finite transport fact follows by sending mass $\mu_-(x)$ from each lower configuration x to upper configurations $y \geq x$ with capacities $\mu_+(y)$. For a set B of lower configurations, the required cut inequality is

$$\mu_-(B) \leq \mu_-(\uparrow B) \leq \mu_+(\uparrow B),$$

so the finite max-flow/min-cut criterion transports all the mass. Apply the random-source assertion to this coupling and multiply by m . The Gibbs continuation changes the resulting difference of source integrals into $\mathbb{E}_\mu[qh]$ and $\mathbb{E}_\mu[qM_J]$ respectively, which proves (4.20). $\square$

Corollary 4.12 (Ordered changes of domains and total variation). *Suppose two ordered Ising laws, possibly defined by different sets of pins or different volumes, have a common free source contour S_0 such that their conditional continuation from S_0 toward a target T is the same. Let an admissible cut J lie between S_0 and T . Then (4.2) holds for their expectation differences. In particular, with total variation defined as the supremum of event probability differences,*

$$\|\mu_T^+ - \mu_T^-\|_{\text{TV}} \leq C(w/s)^{-7/8+\delta} (\mathbb{E}_{\mu^+} M_J - \mathbb{E}_{\mu^-} M_J). \quad (4.21)$$

At fixed positive w/s , convergence of the magnetic difference to zero therefore implies convergence in total variation on the buffered target.

Proof. Restrict an ordered coupling to S_0 and then use the common Gibbs continuation. Corollary 4.11 in its random-source form applies. Taking h to be an arbitrary target-event indicator proves (4.21). All differing pins must be on the source side of S_0 ; the common-continuation hypothesis explicitly excludes a changed pin in the target component. $\square$

5 Nearby contours and prediction

A second consequence of the correlation estimates is that a normalized magnetization can be predicted from a nearby contour. This result supplements ordered transmission when one compares observations on successive cuts; it is independent of the stopping-band construction that follows. We first estimate the difference of two line sums and then use the minimizing property of conditional expectation.

The relative displacement $h = w/s$ is held fixed while $s \rightarrow \infty$. The proof takes the continuum limit of the two-point sums before estimating their change under displacement. It therefore requires no discrete derivative estimate for the two-point function.

Proposition 5.1 (Differences of nearby line sums). *Let J_s, J'_s be two parallel horizontal or vertical lattice segments of lengths $O(s)$ whose matching portions differ by a displacement of length at most Cw and whose unmatched end pieces have total length $O(w)$. The same statement is allowed for two concentric square contours of radii comparable to s and radius difference at most Cw , by matching their four sides. Assume the union has a free neighborhood of width cs and is covered by a fixed number of buffered planar charts. Set*

$$M_{J_s} = s^{-7/8} \sum_{J_s} \sigma_x, \quad M_{J'_s} = s^{-7/8} \sum_{J'_s} \sigma_x.$$

There is a constant C_0 , depending only on the fixed geometry and the buffer, such that, for each fixed $h = w/s \in (0, h_0]$,

$$\limsup_{s \rightarrow \infty} \|M_{J_s} - M_{J'_s}\|_2 \leq C_0 h^{3/8}. \quad (5.1)$$

The law may have arbitrary common pins beyond the buffer or be a mixture of such laws. The same conclusion holds for nearby full periodic rows of period W , normalized by $s^{-7/8}$, when W/s is fixed and positive; its constant may depend on this ratio.

Proof. The pointwise comparison in Corollary 2.5, applied to the nonnegative square of a line difference, reduces each buffered chart to the whole-plane law at the cost of a fixed factor. A finite chart decomposition and the triangle inequality in L^2 will therefore suffice.

In the whole-plane law the mean is zero. By Lemma 3.1, normalized two-point sums on rescaled segments converge to the corresponding integrals of

$$K(x - y) = c_\sigma |x - y|^{-1/4}.$$

If the endpoint data vary with s , first pass to a subsequence realizing the upper limit and then, after a common translation, to a further subsequence on which their rescaled lengths and relative endpoints converge. These data range over a compact set, and all the estimates below are uniform on that set. Here is the required control at the diagonal. The uniform bound $\mathbb{E}(\sigma_x \sigma_y) \leq C(1 \vee |x - y|)^{-1/4}$ shows that parameter pairs whose rescaled distance is at most r contribute at most $C r^{3/4} + o(1)$ to the second moment of a normalized segment sum. The equal-site terms are $O(s^{-3/4})$. Away from the diagonal the two-point asymptotic is uniform, so ordinary Riemann sums converge. These observations justify the integral limit for both self and cross terms, including shifted segments.

For two matching segments, pair their points by a displacement v with $|v| \leq Ch$. In the limiting squared difference, split pairs of their one-dimensional parameters at distance $2Ch$. The near part is bounded by a constant times

$$\int_0^{C'h} r^{-1/4} dr = O(h^{3/4});$$

this bound also applies if one of the shifted kernels has its singularity at a nearby parameter value. For the far part, $|\nabla K(u)| \leq C|u|^{-5/4}$, and the mean value theorem bounds the combined kernel difference in absolute value by $Ch|u|^{-5/4}$. Its integral is at most

$$Ch \int_{2Ch}^{C'} r^{-5/4} dr = O(h^{3/4}).$$

Thus the limiting squared norm of the matching-segment difference is $O(h^{3/4})$. An unmatched end segment of length $O(h)$ has second moment $O(h^{7/4})$ by the same kernel integral, and hence contributes $O(h^{7/8})$ in norm. There are only a bounded number of such pieces. Since $h^{7/8} \leq h^{3/8}$ for $h \leq 1$, the triangle inequality proves (5.1).

For concentric squares, trim corresponding sides to equal lengths, translate their matching portions in the perpendicular direction, and apply the segment result to each side and the short remainders. For full periodic rows, subdivide both rows at the same fixed locations into matching segments of length a sufficiently small fixed fraction of W . Each pair, together with its buffer, lies in a planar chart; if necessary decrease h_0 by a constant depending on W/s . Apply the preceding comparison and sum the finitely many segment norms. Mixtures are permitted since the density comparison is uniform before mixing. $\square$

Corollary 5.2 (Prediction from a nearby contour). *In the setting of Proposition 5.1, let F be a sigma-field containing the spins of J'_s , and write $P_F X = \mathbb{E}[X \mid F]$. Then*

$$\limsup_{s \rightarrow \infty} \|M_{J_s} - P_F M_{J_s}\|_2 \leq C_0(w/s)^{3/8}. \quad (5.2)$$

The normalization on the approximating sum remains $s^{-7/8}$ even when the two contours have different lengths.

Proof. The nearby sum $M_{J'_s}$ is F -measurable. The minimizing property of conditional expectation gives $\|M_{J_s} - P_F M_{J_s}\|_2 \leq \|M_{J_s} - M_{J'_s}\|_2$. Use Proposition 5.1. $\square$

Remark 5.3 (Order of scales). The transmission exponent, the aspect ratios in the local mass construction, and the constants for a fixed family of cuts are chosen first. The bounds are uniform for sufficiently large patch scales. For each of N concentric cuts at spacings comparable to s/N , the local geometry is the same for every N and $w/s \asymp N^{-1}$. Thus the constant in (4.3) is independent of N . For the nearby-line estimate, fix N and then take the mesh sufficiently fine; this threshold may depend on N . A large fixed N may therefore be selected using the transmission and line constants before taking the mesh limit.

6 Complete interface resolutions

The stopping construction uses a fixed face triangulation, whereas the published loop limit uses leftmost interfaces. We must compare complete collections, including the number and orientation of their macroscopic loops: changing one face pairing can merge two loops or split one.

Throughout this section, locators for [2] refer to arXiv version 3. We use its oriented-loop convention in Section 2.5: an Ising circuit is a closed dual path, with no repeated edge, having plus spins on its left and minus spins on its right. It may visit a dual vertex twice. At a checkerboard face there are two noncrossing ways of pairing the two incoming and two outgoing disagreement edges. A *complete resolution* chooses a pairing at every such face and uses every disagreement edge exactly once. Fixing a diagonal in each face and taking the zero set of the affine spin interpolation on its two triangles gives one such resolution, up to a displacement of order δ .

A loop is *leftmost* when the adjacent plus spins along its left side are connected successively by nearest-neighbor steps; this is the strong-path convention of [2, Section 2.5]. The canonical leftmost collection consists of all loops satisfying that convention.

For oriented curves, let d_u denote uniform distance modulo orientation-preserving parametrization; for loops also allow a change of starting point. Distances between loop collections are obtained by matching loops and charging the diameters of unmatched loops, as in [2, Section 2.7].

Lemma 6.1 (Convergence of every complete resolution). *In a plus Jordan domain whose lattice boundary converges uniformly modulo parametrization, every complete resolution of the spin disagreement edges converges to the same oriented nested CLE_3 collection. This assertion is uniform over the choice of the resolution. In particular it applies to a fixed choice of diagonal in every square face.*

Proof. Theorem 6 of [2] gives convergence of the leftmost loops in the collection metric and the following simultaneous approximation. For every $e > 0$, with probability tending to one as $\delta \downarrow 0$, every Ising circuit ℓ of diameter greater than e has a leftmost circuit ℓ^L such that

$$d_u(\ell, \ell^L) \leq e, \quad \text{diam } C < e \quad \text{for every component } C \text{ of } (\ell \cup \ell^L) \setminus (\ell \cap \ell^L).$$

The paragraph following that theorem explains the common-edge content: the shared edges become dense in each macroscopic loop. In particular, above a fixed positive diameter cutoff every circuit shares edges with its matched leftmost circuit, with probability tending to one. These assertions hold for all circuits on the same event, including circuits selected after the spin configuration has been seen.

We explain why this one-sided approximation also supplies coverage and the correct multiplicities for a complete resolution. From any vanishing mesh sequence, first choose errors $e \downarrow 0$ slowly enough that the simultaneous events above have probabilities tending to one. Pass to a subsequence and couple the leftmost collections to their limit so that both their collection-distance error and the all-circuits error tend to zero almost surely. Choose a diameter cutoff which is not the diameter of a limit loop. There are finitely many limit loops above it. They are simple and pairwise disjoint, and distinct members of this finite family have positive mutual distance.

Any two complete resolutions are joined by changing the checkerboard pairings one at a time. Both possible pairings match incoming to outgoing oriented edges. Each change either merges two edge-simple circuits or splits one into two edge-simple circuits. Consequently all intermediate circuits remain in the class covered by the all-circuits assertion; no crossing circuit or repeated edge is introduced.

Suppose, along a sequence of finer meshes, that some such switch changes the number of circuits matched near a fixed positive-diameter limit loop. Select one offending switch on each mesh. The all-circuits assertion holds for these selected circuits, even though the selection depends on the configuration. After passing to a subsequence, each constituent either has vanishing diameter or converges, modulo parametrization, to a simple loop in the limiting collection. Two vanishing constituents cannot merge to macroscopic size because they meet at the switched vertex. Adding a vanishing constituent to a large one does not change the latter's parametrized limit. If both constituents are large, their meeting vertices force their limit loops to meet. The limits must therefore be the same simple loop.

The last alternative is impossible. Choose a point strictly inside that simple loop. Uniform parametrized convergence preserves the winding number about the point. Each constituent has winding number 1 or -1 , whereas the merged traversal has their sum, namely 0 or ± 2 . The merged circuit is itself covered by the simultaneous assertion. Its image approaches the same loop, so uniqueness of the nearby limiting loop makes its *oriented parametrized* limit that loop as well. Its winding number must therefore be ± 1 , a contradiction. Splits are the reverse operation. It follows that no switch can change the macroscopic count near any fixed limit loop, for all sufficiently fine meshes on this coupled subsequence. This argument selects an offending switch if one exists and therefore does not accumulate an error over the number of switches.

A specified leftmost circuit can be included in a complete resolution: fix its pairings at every visited vertex and resolve the unused edges arbitrarily. At a once-visited checkerboard vertex the unused incoming and outgoing edges have the remaining noncrossing pairing. At a twice-visited vertex the circuit already specifies both pairs. Thus its oriented noncrossing traversal gives consistent local choices. In this resolution a second circuit matched to the same limit loop would have to share edges with the specified leftmost circuit. Uniqueness of the nearby macroscopic leftmost loop and the common-edge consequence above give this conclusion. It contradicts use of every edge only once. The count is thus one in this resolution, and the switching argument makes it one in every resolution. The all-circuits assertion also excludes unmatched positive-diameter circuits. This proves convergence in the collection metric.

Finally, colors are recovered by nesting from exterior plus. Any ancestor of a loop of diameter at least $a > 0$ also has diameter at least a . There are only finitely many such ancestors. Their

nesting and orientations are stable under convergence to disjoint simple loops, so the colors needed at positive scales converge as well. $\square$

7 Stopped interfaces and signed slit data

We now turn the complete loop limit into the boundary information needed for a conditional target law. The exploration retains its full spin transcript, while the geometric datum records only the boundary visible from the target. The next section will replace that boundary uniformly over nearby transcripts. All distances in this and the next two sections are macroscopic: the lattice is $\delta\mathbb{Z}^2$, with $\delta \downarrow 0$.

7.1 Band geometry and the information revealed

Write $Q_r = \{z \in \mathbb{R}^2 : \|z\|_\infty < r\}$ and $\gamma_r = \partial Q_r$. Fix $0 < u < v$ and positive clearances from the band $\overline{Q_v} \setminus Q_u$ to the target and to the boundary of the ambient chart. There are two versions of the construction.

1. In the inward version, initially reveal the spins outside Q_v , and explore from γ_v towards γ_u . The target is a fixed compact subset of Q_u ; in applications it is contained in a strictly smaller concentric square.
2. In the outward version, initially reveal the spins in Q_u , and explore towards γ_v . There is a more distant fixed square wall, and the target lies between γ_v and that wall, with positive clearances from both. The distant wall has prescribed signs, including the choice of a constant minus wall.

The statement also allows a compact interval of choices for each radius, provided the intervals are separated and all the stated clearances hold uniformly. For a requested radius r , choose deterministically a nearest $r_\delta \in \delta\mathbb{N}$, with ties resolved by a fixed rule, and use the contour ∂Q_{r_δ} . For sufficiently small δ these rounded radii preserve all the prescribed separations, and their contours follow primal grid edges. The interpolated zero set meets such a contour only at edge midpoints: it never passes through a spin vertex, and the adjacent triangles lie on opposite sides of the contour edge. Thus these meetings are crossings, including near the square corners. In either version include the vertices of the rounded starting contour in the initial revealed layer. For measurability statements, the *band spins* include the vertices of triangles meeting the closed rounded band. This convention adds only an $O(\delta)$ layer and includes the starting-contour vertices.

Remark 7.1 (Thin fixed collars). No universal lower bound on $v/u - 1$ is required. Each collar has strictly positive width fixed before the mesh limit, and the constants and mesh threshold may depend on that width. For example, collinear segments of length ℓ , separated by gaps $g > 0$, fit in disjoint squares with containing halfside $\ell/2 + g/8$ and farther-wall halfside $\ell/2 + g/4$. There is room between these squares for any fixed finite list of nested contour levels with positive clearances. When pieces come from a polygonal contour, first remove neighborhoods of corners and boundary intersections, then take segment lengths smaller than those separations. The remaining straight pieces admit the same construction. Its constants can depend on g/ℓ ; a fixed enlargement factor independent of g/ℓ is unnecessary.

When radii vary, reserve one fixed larger permitted window containing their compact ranges, and choose all common source and test contours with positive clearance from the entire range of possible stopping bands. The regular barriers constructed below stay on the fresh side of the realized

visible wall, with the prescribed target clearance; they need not lie between each realization's numerical radii u, v . The reserved window provides room for their offsets and tip caps.

We encode lattice boundary data on the requested continuum contours. Choose disjoint fixed intervals about the two permitted radius ranges, with positive endpoint clearances. Let $f_\delta : [0, \infty) \rightarrow [0, \infty)$ be the increasing piecewise-linear map that sends u_δ to u and v_δ to v , fixes the interval endpoints, and is the identity outside those intervals, linear between consecutive specified knots. Define

$$R_\delta(0) = 0, \quad R_\delta(z) = \frac{f_\delta(\|z\|_\infty)}{\|z\|_\infty} z \quad (z \neq 0).$$

This homeomorphism and its inverse move every point by $O(\delta)$, uniformly over the permitted radii: on each affine interval the error $f_\delta(r) - r$ interpolates endpoint errors of size at most $\delta/2$. It maps the closed rounded band onto the exact band and each rounded contour onto its requested counterpart. We apply R_δ only when recording the geometry of the exploration below. The physical spins, reveal, and Gibbs laws remain on the original lattice.

For the probability limit, sample spins in a fixed larger square with plus boundary conditions. Other buffered laws will enter only through Corollary 2.5. The sign-reversed reference square may equally well be used. A finite collection of band observations is always computed from one common spin configuration, so that its joint law has an unambiguous meaning.

Fix one diagonal in every square face, without adding interactions to the Ising Hamiltonian. Affinely interpolate the spins in each triangle. The zero set consists of noncrossing polygonal strands. After the initial reveal, inspect every strand meeting the starting contour. Follow it on the fresh side until its first return to the starting contour or its first hit of the other contour. In the latter case also follow the strand from its other starting endpoint, when there is one, stopping by the same rule. Every time a triangle is traversed, inspect its vertex spins. The order of exploring the starting endpoints is fixed deterministically. Repeated inspections reveal no additional information.

From the target, the visible boundary consists of a base formed by the starting contour and returning excursions, together with strands stopped at the second contour and their two side signs. This is the target-facing datum formalized in Definition 7.3.

Lemma 7.2 (Stopping property). *For each $\delta > 0$, this is a finite adaptive spin reveal. Conditional on its complete information, the unexamined spins have the Ising Gibbs law with the examined spins fixed. The revealed set lies on the starting side and in an $O(\delta)$ enlargement of the band. The traced strands, their side signs, and the target-facing boundary datum defined below can be computed from the band spins alone.*

Proof. There are finitely many starting edges and finitely many triangles. At a step, the next triangle is determined by the previously exposed zero segment and the fixed diagonal convention. Its spins determine the continuation. Hence the event that a specified reveal transcript occurs depends only on the spins recorded in that transcript. In the finite Gibbs density, conditioning on such a transcript fixes precisely those spins and leaves the original interaction weights on all remaining variables. This proves the stopping assertion directly, including the case of a random examined set.

The trace stops at the second contour and examines only adjacent triangles, which proves the location assertion. Although the initial reveal contains an entire side of the band, the algorithm deciding a strand within the band uses only the spins adjacent to the starting contour and in the band itself. The target-facing datum is determined by these traces and their local signs. No spins in the rest of the initially revealed side are needed to compute it. $\square$

7.2 The visible Jordan base and its slits

The following definitions refer first to an oriented collection of simple disjoint continuum loops. Restrict a loop to the fresh side of the starting contour. A maximal such portion returning to that contour before reaching the second contour is a *returning chord*. A portion reaching the second contour is retained only up to its first hit there and is a *stopped strand*. Cut the fresh region along these curves and take the component containing the untraced region beyond the second contour. Retain only boundary pieces seen from this component. The same definition is made for the resolved lattice walls.

The boundary is read from the remaining domain: a traversal follows one side of each slit to its tip and returns along the other. Thus the two sides of a slit are distinct boundary arcs even though they have the same geometric image. This is the prime-boundary traversal used below.

Definition 7.3 (Admissible facing datum). A facing datum consists of an oriented Jordan base, finitely many pairwise disjoint simple slits attached to it at distinct roots, the root and tip marks, and signs on the two sides of each slit and on the intervening base arcs. Signs on a slit are opposite. In the prime-boundary traversal the signs change at its tip and continue unchanged through each occurrence of its root. The whole datum lies in the closed band, and its tips lie on the second contour. If there are no slits, the base has one constant side sign.

The lattice datum is the image of its physical boundary record under R_δ , retaining roots, tips, orientation, and side signs. When radii vary over permitted windows, $\mathcal{X}_{\text{band}}$ denotes the union of admissible data from those radius choices. The numerical radii are not extra marks; for example, a datum without slits need not determine its second radius.

Within a fixed finite combinatorial type, use the maximum uniform distance modulo increasing parametrizations of the base and slits, respecting root attachments, side labels, and orientation. Cap this distance at one and put distance one between different types. Denote this metric space by $\mathcal{X}_{\text{band}}$.

For approximation centers we also use an ambient space $\hat{\mathcal{X}}$ of finite marked continuous walks of these combinatorial types. Retain the orientations, side labels, root and tip marks, and shared coordinates of attached roots, but drop simplicity, disjointness, and the exact band-containment and contour-incidence requirements. Use the same matching-distance formula, identifying representatives at distance zero. Then $\mathcal{X}_{\text{band}} \subset \hat{\mathcal{X}}$ has its stated metric as the induced metric. A ball with an ambient prototype as center always means its intersection with $\mathcal{X}_{\text{band}}$ when used as a set of facing data.

Lemma 7.4 (Visible data and their continuity). *For almost every choice of the two contour radii, the continuum construction from nested CLE_3 yields an admissible facing datum. Its base may contain infinitely many returning chords but is a Jordan curve. The lattice data converge to it in $\mathcal{X}_{\text{band}}$. These assertions hold jointly for any fixed finite collection of bands.*

Proof. We give the deterministic continuity argument on the probability-one event of simple, disjoint loops and local finiteness at every positive diameter. For a continuous real function on a compact parameter interval, its local maximum and minimum values form a countable set: each is the maximum or minimum on an interval with rational endpoints contained in a witnessing neighborhood. Apply this to the square radius along each loop. There are countably many loops, so the union of exceptional levels is countable for each collection. Fubini's theorem permits a deterministic full-measure set of radius choices for the random collection. At the selected levels there is no one-sided contact or constant interval.

Only finitely many excursions have diameter greater than a prescribed $a > 0$. Indeed, only finitely many loops have that diameter; on each, uniform continuity of a parametrization gives a

positive minimum parameter length for such an excursion. Their parameter intervals are disjoint. A stopped strand has diameter at least the positive distance between the contours, so there are finitely many stopped strands. Their tips are distinct. The two stopped ends of the same deep excursion could have the same tip only if that tip were a one-sided contact at the second level, which was excluded.

Distinct excursion endpoints on the starting contour are also distinct. Endpoints on different loops cannot coincide by disjointness. A shared endpoint for two excursions of one simple loop would give a one-sided level contact. Excursion interiors miss the starting contour, and simplicity and disjointness exclude all other intersections.

Each visible returning chord deletes the open interval of the starting contour which is cut off from the target component. Visible deletion intervals have disjoint interiors: in a pair of nested deletions the inner chord is screened and is not visible. A small chord deletes a small interval. To see this, join its endpoints by the short arc of the starting square. As the chord diameter tends to zero, uniform continuity of the inverse parametrization of that square makes this arc small. The resulting small Jordan pocket cannot contain the target beyond the second contour, which is separated by a fixed positive distance. The erased interval must therefore be this short arc, not the complementary long arc. This argument works also on the exterior side; equivalently, compactify the exterior by a fixed inversion away from the band.

Parametrize the starting contour and replace every deleted interval by a parametrization of its chord with the same endpoints. Retain the original parametrization elsewhere. At an accumulation of replacement intervals, the interval diameters and chord diameters tend to zero, by the preceding finiteness and small-pocket facts. The resulting map is consequently continuous. It is injective: chord interiors miss the starting contour and one another, their endpoints are distinct, and retained points of the original contour remain distinct. A continuous injection of the compact circle into the plane is a Jordan parametrization. This proves, in particular, that infinitely many small returning chords do not create an accumulating pinch. Adding the finitely many visible stopped strands gives the asserted base and slits.

For continuity, first choose $a > 0$ outside the countable set of excursion diameters and retain the finitely many excursions of diameter greater than a . Uniform convergence of the parent loops gives convergence of these excursions, their endpoints, and their first hits of the second contour. Here is the role of the generic levels. A compact subarc strictly on one side of a level stays there under a small uniform perturbation. Arbitrarily near an excursion endpoint on its other side in parameter time, the limiting loop has points on the opposite side of the level; otherwise the endpoint would be a one-sided contact. These points bracket the perturbed endpoint between arbitrarily nearby parameter values. The same argument at the cutoff brackets its first hit. Splitting, merging, or losing a retained excursion would contradict one of these bracketing properties.

The circular order of the finitely many retained endpoints is stable. Their visibility is stable as well. An additional chord of small diameter cannot screen a retained chord: its deleted pocket has small diameter and cannot contain that retained chord. After matching the retained pieces, match the remaining intervals of the bases in their circular order. All discrepancies on these intervals have diameter tending to zero as $a \downarrow 0$. This constructs parametrizations with uniformly small distance. Include the finitely many slit roots in the matching; their stopped strands already converge by the same bracketing argument. In particular the fixed band width and the generic second level give eventual equality of the finite stopped-strand list, with its cyclic root and tip order. Only the returning chords require the subsequent limit $a \downarrow 0$. The recording map R_δ adds an error of $O(\delta)$ and ensures that the lattice records have the exact contour incidences required by the data space.

At the lattice level, a returning chord replaces an interval whose two surviving side signs agree.

The only sign changes in a traversal of the visible prime boundary occur at stopped tips. Thus microscopic returning chords introduce no additional alternations between tips. When slits are present, their oriented sides determine the signs on all intervening base arcs, so these signs converge by Lemma 6.1.

To recover the constant sign when there are no tips, we use one further property of nested CLE_3 : a deterministic interior point is almost surely surrounded by infinitely many nested loops [14, Section 1, before Proposition 1]. Their diameters tend to zero by local finiteness. Choose such a point on the starting contour. A sufficiently small surrounding loop misses the second contour and has a returning excursion into the fresh side. If this chord is hidden, any chord screening it has diameter bounded away from zero by the small-pocket argument. There are only finitely many such screening chords, so one is visible. Its oriented parent loop supplies the constant side sign. That loop and its finitely many ancestors converge with their orientations by Lemma 6.1. This proves sign convergence also in the no-slit case. The additional fixed-point nesting property is used only for this color anchor, not for the preceding deterministic construction of the Jordan base.

The deterministic argument applies simultaneously to any fixed finite list. Lemma 6.1 and the continuous mapping theorem finish the proof. $\square$

Remark 7.5. The ambient space $\hat{\mathcal{X}}$ in Definition 7.3 is separable. Approximate finitely many parametrized walks by rational piecewise-linear walks, approximating shared endpoints and roots together. These prototypes need not be simple or satisfy the exact band constraints; they form a countable dense set in the ambient matching metric. The countable union over finite combinatorial types remains separable. An ambient countable base restricts to a countable base on $\mathcal{X}_{\text{band}}$, so that subspace is also separable. Only realizable admissible data, not the prototypes, are used to construct Gibbs boundaries.

8 Regular boundaries and ordered erosion

The visible boundary from Section 7 is a Jordan base with finitely many signed slits. We now replace it by one deterministic regular boundary. The replacement must work for every sufficiently nearby realizable stopping transcript, even though those transcripts can have different examined sets. This uniform local statement is what will permit the finite probability cover in Section 9.

8.1 Local replacement and the reduction to spin means

Choose a source contour S_0 and a square test contour J in the free region between the entire permitted stopping band and the target. Choose their positive collars so that J is an admissible cut between S_0 and the target, in the sense of Definition 4.1. All modifications of the facing boundary occur on the band side of S_0 . For each fixed distant-wall assignment, the two laws in an ordered insertion therefore have the same conditional continuation from S_0 toward the target. Write

$$M_J^\delta = \delta^{7/8} \sum_{x \in J^\delta} \sigma_x. \quad (8.1)$$

Lemma 3.3 bounds its second moment uniformly. The fixed macroscopic reference scale implicit in Corollary 4.12 only changes its constant.

Lemma 8.1 (Local ordered erosion). *Fix an admissible facing datum d , the band geometry, a target beyond the band with a free collar, and $\varepsilon > 0$. There exist a neighborhood U_d of d in $\mathcal{X}_{\text{band}}$, a*

regular Jordan contour Γ_d on the fresh side of the datum with finitely many signed arcs, and $\delta_d > 0$ with the following property.

For every $0 < \delta < \delta_d$ and every realizable stopping transcript whose datum belongs to U_d , the lattice boundary layer following Γ_d is free in the original conditional law. Let μ^0 be that law, let μ^+ be obtained by inserting its prescribed plus pins, and let μ^f be obtained by then inserting its prescribed minus pins. The two changes are ordered, and

$$\|\mu^0|_{\text{target}} - \mu^+|_{\text{target}}\|_{\text{TV}} + \|\mu^+|_{\text{target}} - \mu^f|_{\text{target}}\|_{\text{TV}} < \varepsilon. \quad (8.2)$$

On the fresh side of the full contour, μ^f is exactly the Gibbs law determined by this contour and, in the outward case, the unchanged distant wall. In particular it depends on the chosen contour, the mesh, and the unchanged distant wall, and not on the transcript within U_d .

The contour can be chosen polygonal or smooth, with its sign switches on smooth portions. Its positive clearances, the neighborhood, and the mesh threshold may depend on d and ε , but not on the remaining conditional spins or on the chosen transcript.

To see what the construction must establish, consider any candidate barrier on the band side of S_0 whose layer is free in the original law, and use the three laws in the statement. Put $m_j = \mathbb{E}_{\mu_j} M_j^\delta$ for $j \in \{0, +, f\}$. Inserting plus pins is upward and inserting minus pins afterward is downward, so $m_+ \geq m_0, m_f$. Corollary 4.12 gives

$$\begin{aligned} \|\mu^0|_{\text{target}} - \mu^+|_{\text{target}}\|_{\text{TV}} + \|\mu^+|_{\text{target}} - \mu^f|_{\text{target}}\|_{\text{TV}} \\ \leq C[(m_+ - m_0) + (m_+ - m_f)]. \end{aligned} \quad (8.3)$$

The constant depends only on the fixed free collars.

We will construct two deterministic regular signed boundaries, one upper and one lower, that approximate the same slit datum. Write m_u and m_l for their test-contour means. After changing pins only in small neighborhoods of the finitely many roots and tips, boundary monotonicity will place every nearby transcript law between these two regular laws. If the resulting mean error is at most e , this gives

$$m_l - e \leq m_j \leq m_u + e \quad (j = 0, +, f),$$

and the right side of (8.3) is at most $2C(|m_u - m_l| + 2e)$. The small-patch comparison makes e small. Marked-domain convergence makes the two regular means approach the same limit.

These are separate requirements. The next subsection proves the marked convergence for deterministic ribbon outlines, keeping the two sides of each slit distinct. The incidence criterion then proves the ordering for actual examined triangles. Their combination will control every transcript in one neighborhood without requiring a spin-scaling limit in its irregular domain. For an outward band the analytic argument first uses a constant distant wall; the common-weight lemma below restores arbitrary prescribed wall signs with a uniform change in the error budget.

8.2 Regular outlines and marked convergence

An admissible datum bounds a disk with finitely many attached slits in the inward case. In the outward case it bounds an annulus with slits attached to its inner boundary; the outer wall is unchanged. Opposite signs on the two sides of a slit refer to its prime-boundary sides. They do not prescribe two signs to a single lattice vertex: actual Gibbs boundaries will be narrow outlines with separate sides.

Lemma 8.2 (A chart for a finite slit graph). *Let a Jordan base have finitely many disjoint attached simple arcs at distinct roots. There is a homeomorphism on a neighborhood of this graph which makes the base a regular circular line and makes the arcs disjoint straight segments on its fresh side. The chart may be chosen with a further open collar beyond every tip. A fixed compact target disjoint from the graph remains outside a sufficiently small neighborhood used for this construction.*

Proof. Subdivide the base at its roots and add further vertices as needed so that the base and attached arcs form a finite simple plane graph. Theorem 1 of [13], with “isomorphic” defined there by a homeomorphism of the plane, gives a polygonal embedding obtained by an ambient homeomorphism. In particular, the free tip of an attached arc has a neighborhood in which it is a straight half-edge and can be prolonged a short distance in the complement. This supplies no common metric modulus. The Jordan–Schoenflies plane extension theorem [11, Section 10, Theorem 4] also supplies a collar of the base. We describe the compatibility construction for all the arcs. Prolong each slit from its free tip to a surrounding curve on the fresh side, choosing the prolongations disjoint. This can be done successively: an arc attached at just one endpoint to a boundary does not separate the adjacent open region, so its free endpoint can be connected to the surrounding curve without crossing the other attached arcs. A connecting path can be simplified to a simple arc. After one prolongation has been chosen, cut along it and repeat in the appropriate Jordan region. Choose distinct endpoints on the surrounding curve in the cyclic order of the roots. This preserves disjointness at each step.

The prolonged arcs cut the collar into finitely many Jordan cells. Map their boundary arcs, compatibly at common edges, to the sides of rectangular cells in a circular strip, sending each marked tip to an interior point of the corresponding straight side. Apply the same plane extension theorem in each cell and glue the extensions. The glued map and its inverse are continuous because there are finitely many closed cells and their boundary maps agree. This gives the required chart. When there are no slits the initial collar suffices. The surrounding curve and all prolongations can be chosen in a neighborhood of the finite graph disjoint from the target; equivalently take a sufficiently small regular neighborhood after the successive extensions. Compactness and the positive distance to the target provide the required final clearance. $\square$

Hereafter a displacement parameter such as q or ρ is measured in this chart. We do not assume that the chart is Lipschitz. Once finitely many chart parameters have been fixed, disjoint compact pieces have positive physical separation. The tolerance for perturbing data and the lattice mesh will be chosen below that separation. For a lattice transcript, undoing the recording map R_δ from Section 7 moves the curves by $O(\delta)$ in the physical plane. Thus a small recorded-data tolerance, followed by a fine mesh, gives any prescribed chart tolerance for the actual wall. All separation and incidence statements below concern that physical wall and absorb this displacement into their positive slack.

Lemma 8.3 (Small regularization of an outline). *A Jordan outline, finitely many marked points on it, and a sufficiently small surrounding Jordan collar admit a polygonal or smooth replacement through an ambient homeomorphism supported in that collar. Its displacement can be made arbitrarily small. The marked points retain their cyclic order and can be placed on straight grid-aligned portions of the replacement. Finitely many disjoint outlines can be treated simultaneously.*

Proof. Use a Schoenflies parametrization of a closed annular collar of the outline. Uniform continuity permits a sufficiently thin collar and a cyclic subdivision into small quadrilateral strips, with small open overlaps between consecutive strips and with nonconsecutive strips disjoint. Take marked points away from the overlaps. In each open overlap choose a joining point. Join successive points

by polygonal arcs in the intervening open strip, erasing loops. Within an overlap, truncate the two neighboring arcs before their first intersection and join them by a simple polygonal bridge. The overlaps are disjoint, so this operation is local. This gives a Jordan polygon following the prescribed cyclic order. All polygonal endpoints used in this construction are in open sets; no straight approach to an arbitrary point of a rough cell boundary is required. At a marked strip insert a short horizontal or vertical segment and put its mark there. Smooth corners, if desired, away from the marks.

Choose transverse crosscuts in the small overlaps to subdivide the old and new collars, respectively. They give corresponding Jordan cells above and below the old outline and the new polygon. Their outer boundary maps are the identity; on their middle boundaries map the old arc onto the new arc and match the marked points. Choose agreeing maps on the corresponding transverse crosscuts. The plane extension theorem extends these maps cell by cell. The extensions glue to a homeomorphism equal to the identity outside the collar. Corresponding old and new cells lie in the same small strip and its adjacent overlaps, so displacement is bounded by that neighborhood's diameter. These diameters were made arbitrarily small. The inverse has the same displacement bound. Disjoint collars permit the finite simultaneous construction. $\square$

For the next lemma, normalize a disk uniformizer by $F(0) = z_*$ and $F'(0) > 0$, where z_* is a fixed interior point. For an annulus, preserve the inner and outer labels and normalize the inverse map by $g(z_*) > 0$. Marked convergence additionally tracks the preimages of the finitely many sign switches on the boundary circles.

Lemma 8.4 (Marked convergence of ribbon outlines). *Let D be a disk or annulus bounded on one side by an admissible slit datum and, in the annular case, on the other side by a fixed disjoint Jordan wall carrying a fixed finite collection of nondegenerate signed arcs. Replace each slit by a ribbon, with width and displacement tending to zero in the chart of Lemma 8.2, and offset the base by a quantity tending to zero. Use straight strips away from nodes and fixed polygonal templates, scaled to the offset size, at the finitely many nodes. The templates are chosen with no narrowing neck beyond a fixed fraction of their scale. Approximate the resulting outlines by smooth or polygonal Jordan curves through homeomorphisms whose physical displacement tends to zero, as in Lemma 8.3. Preserve boundary-component labels and the cyclic order of the side signs. Put one sign-switch mark on each cap, tending in the chart to its designated slit tip. One may also retain finitely many additional sign-switch marks on the base away from slit roots, tracking them by their boundary parametrization.*

Then the resulting domains D_q converge to D in the marked Carathéodory sense. Normalized uniformizing maps converge uniformly up to the boundary, after radial identification of their source annuli when necessary. Every cap mark converges to the prime end at its slit tip. If m_q and m are the corresponding continuum critical Ising one-point functions, then, for every $K \Subset D$,

$$\sup_{z \in K} |m_q(z) - m(z)| \longrightarrow 0. \quad (8.4)$$

Proof. We spell out the boundary control because unmarked kernel convergence alone would not identify the cap marks. It suffices to prove the assertion along an arbitrary null sequence of positive parameters $q_n \rightarrow 0$. In the proof all statements uniform in q concern this sequence, with its index suppressed.

Connectivity of the filled complements. In the straightened chart the filled complementary component is a base region with finitely many thin rectangular teeth. Two nearby points in one straight tooth can be joined by a segment or a two-segment path of diameter at most a fixed

multiple of their distance. The same is true in the base region. Near a junction, the finitely many fixed polygonal templates have a finite local quasiconvexity constant, which is unchanged on scaling. Nonincident pieces away from their junctions have fixed positive separation. Consequently there are $C, t_0 > 0$, independent of q , such that points at distance $t < t_0$ in the same model filled complementary component can be joined there by an arc of diameter at most Ct .

Let H be the fixed homeomorphism from the model to the physical chart. Write the polygonalization or smoothing map as J_q , so the physical complementary components are images under $H_q = J_q \circ H$. The maps J_q and J_q^{-1} tend uniformly to the identity. Thus H_q and H_q^{-1} have common moduli of continuity μ and ν on the relevant compact neighborhoods: uniform convergence controls the tail of the sequence, and continuity controls its finite head. Two physical points at distance t in the same complementary component can be joined there by an arc of diameter at most

$$\theta(t) = \mu(C\nu(t)), \quad \theta(t) \longrightarrow 0. \quad (8.5)$$

The fixed other complementary component in the annular case has the same property by its Jordan parametrization, enlarging θ if necessary. We require this property for the filled components, not for their boundary curves; the latter need not be uniformly locally connected when a ribbon narrows.

Interior uniformization. For the classical disk and annulus uniformization theorems, including uniqueness of the annular radius ratio, see [12, Theorems 1.2.1 and 1.3.1]. The labeled complementary components converge in Hausdorff distance, and compact subsets of D are eventually contained in D_q . The normalized kernel theorem gives interior convergence for disks [3, Definition 2.8 and Theorem 2.9]. Use the modulus convention $m(A(r, R)) = (2\pi)^{-1} \log(R/r)$. For annuli, write the source annulus as $A(r_q, 1)$, preserving the boundary-component labels; existence of these uniformizers follows from [12, Theorem 1.3.1]. A fixed smooth essential annulus compactly contained in D is eventually contained in every D_q . Also every D_q is contained in a fixed bounded annulus whose inner disk lies in the inner complementary component. Both inclusions are essential: the intervening annuli separate the same two complementary components. Monotonicity of annular modulus [1, Proposition 4.3.5 and Section 4.3.4] therefore gives

$$0 < r_- \leq r_q \leq r_+ < 1.$$

To identify a subsequential limit $r_q \rightarrow r \in (0, 1)$, write the inverse uniformizers as $g_q : D_q \rightarrow A(r_q, 1)$ and the forward maps as $F_q = g_q^{-1}$, with the inverse normalized at a fixed $z_* \in D$. Take simultaneous locally uniform normal limits g on D and F on $A(r, 1)$. The inverse maps have winding number one on a fixed essential curve and are bounded away from zero. Their limit cannot be constant, since a nonzero constant has winding number zero. The maximum and minimum principles give $r < |g| < 1$ on D . Consequently local convergence in the identity $F_q(g_q(z)) = z$ gives $F(g(z)) = z$ for every $z \in D$, so F is nonconstant.

We must also exclude values on the limiting slit boundary. For every finite $a \in \mathbb{C} \setminus D$, Hausdorff convergence of the labeled filled complementary components supplies $a_q \in \mathbb{C} \setminus D_q$ with $a_q \rightarrow a$. The functions $F_q - a_q$ are zero-free. Hurwitz's Theorem and the nonconstancy of F imply $F(w) \neq a$ for every $w \in A(r, 1)$. Thus $F(A(r, 1)) \subset D$. Applying local convergence now to $g_q(F_q(w)) = w$ gives $g(F(w)) = w$. The two limits are therefore conformal inverses onto their domains. The modulus of D is unique, which identifies r ; normalization identifies the maps. Every subsequence has these same limits, proving interior convergence of the moduli and maps.

Boundary equicontinuity. Let F_q denote a forward map from the unit disk or the source annulus. Disk uniformization and the Jordan boundary-extension theorem are recalled in [12, Theorems 1.2.1–1.2.2]. For each positive q the annular target boundaries are also Jordan; the annular extension preserving boundary-component labels is [12, Theorem 1.3.2]. Thus each map

extends continuously to its boundary. Its image lies in a fixed bounded set, and univalence gives a uniform area bound

$$\int |F'_q(z)|^2 dA(z) \leq A_0.$$

Fix a source boundary point ζ . Choose $R > 0$ uniformly smaller than the source collar width and the distance of the source normalizing point from the boundary. For $0 < r < R$, the circular arcs $C_t = \{|z - \zeta| = t\}$ in the source domain, $r < t < R$, are crosscuts with both endpoints on the same boundary component. If $\ell_q(t)$ is the length of $F_q(C_t)$, Cauchy–Schwarz and polar integration give

$$\int_r^R \frac{\ell_q(t)^2}{t} dt \leq 2\pi A_0.$$

For some $t \in (r, R)$ it follows that

$$\ell_q(t) \leq s(r) := \left(\frac{2\pi A_0}{\log(R/r)} \right)^{1/2}. \quad (8.6)$$

The image endpoints can be joined in their common filled complementary component by a simple arc of diameter at most $\theta(s(r))$. This arc meets the image crosscut only at its endpoints. Their union is a Jordan curve in a ball of radius $s(r) + \theta(s(r))$ about one endpoint. For small r the ball excludes z_* , whose distance from all target boundaries is bounded below. In the annular case take it also smaller than the separation of the complementary components. The source cap cut off by C_t does not contain the source normalizing point. Its image must be on the small bounded side of this Jordan curve: the other component contains z_* , and the curve meets the domain only in the crosscut. Therefore

$$\text{diam } F_q(B(\zeta, r) \cap \text{source domain}) \leq 2[s(r) + \theta(s(r))] \longrightarrow 0 \quad (8.7)$$

uniformly in q and ζ . Interior convergence and this estimate prove uniform convergence on the closures. For annuli use the radial maps between $A(r, 1)$ and $A(r_q, 1)$; since $r_q \rightarrow r \in (0, 1)$ these maps and their inverses have uniformly bounded Lipschitz constants and tend uniformly to the identity.

Marks. A tip of a finite simple slit has exactly one prime end above it. In the chart, a small disk about a tip is cut along a segment ending there; nested crosscuts around the endpoint give its unique approach class. A root can have two preimages in the prime boundary, but no sign change is placed at a root. If b_q is a cap mark, any limit of $F_q^{-1}(b_q)$ maps to the designated tip by uniform convergence on the closures. Uniqueness of its prime-end preimage identifies the limit. An additional mark on the base away from the roots has the same uniqueness property because the base is Jordan and no slit is incident there. Distinct prescribed marks give distinct limiting prime ends, so all sign arcs remain nondegenerate.

Spin means. Theorem 1.3 of [4], with the convergence convention of its Definitions 3.1–3.2, applies to the fixed smooth or polygonal domains here and to the limiting marked disk or annulus. For a slit domain, boundary arcs are interpreted on the prime boundary through the conformal pullback in its Definition 3.6. The preceding mark argument preserves the number of boundary components, the cyclic arc order, and the nondegeneracy of every sign arc. The rough-boundary local normalizations of its Lemma 3.34 cancel in the correlation ratios in the proof of Theorem 1.3. Its bulk convergence is uniform on compact sets. Continuity of the continuum correlations under the marked convergence above follows directly by a diagonal argument: if Equation (8.4) failed, choose an offending sequence of continuum outlines and then a lattice approximation to each so fine

that its normalized mean is within the reciprocal of the sequence index of that continuum mean. Choose the approximations also sufficiently close in the marked topology. The combined lattice sequence approximates the limiting marked domain, contradicting the same bulk convergence theorem. This proves Equation (8.4), including uniformity over K . $\square$

8.3 The boundary-incidence criterion

Marked convergence controls the means for the regular comparison outlines. To compare an irregular realized wall with these outlines, we still need a discrete ordering valid for every nearby transcript. The next two lemmas provide that ordering from the signs encountered by the common target component.

We next separate the discrete ordering statement from its geometric verification. All models are represented on a common finite nearest-neighbor graph by declaring some vertices pinned. Vertices beyond a complete boundary layer can be assigned arbitrarily, because that layer separates them from the target.

Lemma 8.5 (Ordering by boundary incidences). *Let μ_0, μ_1 be two ferromagnetic Ising laws with the same interactions and possibly different pin sets. Let C be the component of the vertices free in both models which contains a specified connected target. Suppose that, for every edge xy with $x \in C$ and $y \notin C$, at least one of the following holds:*

1. y is pinned minus in model 0;
2. y is pinned plus in model 1;
3. y is pinned to the same sign in both models.

Then $\mu_0|_C \preceq \mu_1|_C$. The same assertion holds for a union of target components with the corresponding incidence condition.

Proof. Condition separately on all spins outside C . A boundary neighbor cannot be free in both models, since its edge to C would put it in that common free component. Each of the three alternatives gives $\sigma_0(y) \leq \sigma_1(y)$ for every pair of exterior assignments consistent with the pins. Thus at each $x \in C$ the induced boundary field in model 0 is at most that in model 1. Their internal couplings agree and are ferromagnetic. A monotone coupling of the finite heat-bath chains on C , or equivalently the ferromagnetic boundary monotonicity, gives domination for each pair of exterior assignments. Mixing them proves the marginal assertion. $\square$

Lemma 8.6 (Examined triangles preserve the facing sign). *Fix an admissible limiting datum and disjoint small neighborhoods of its roots and tips. For sufficiently close resolved data and sufficiently fine mesh, an edge from a free vertex in the target component to an examined neighbor, outside those neighborhoods, sees the prescribed sign of the corresponding accessible face of the ideal datum. The short approach to the realized wall has length $O(\delta)$ in the physical plane. Its identification with a limiting wall face also uses the datum tolerance and the fixed chart's modulus of continuity.*

Proof. An edge between free vertices cannot cross a traced zero segment: the traversed triangle containing such a crossing would have caused both endpoints to be inspected. The same observation applies to the open part of an edge whose last endpoint is examined. If that endpoint was inspected in a traversed triangle, append an approach of length $O(\delta)$ in that triangle, on the side of the zero segment carrying its spin sign. Affine interpolation gives a constant vertex sign on each side of that segment. If instead it belongs to the initial revealed side, the edge lies within $O(\delta)$ of the starting

polygon; approach the accessible arc of that polygon directly. Every initial sign switch produces a zero crossing of the starting contour and is explored by the algorithm. Hence no additional switch is present on a retained base arc away from its roots. In either case this constructs an approach to the realized wall from the target-accessible side, carrying the examined endpoint's sign.

Outside the node neighborhoods, use the chart of Lemma 8.2. The limiting edge lies in a strip disjoint from all nonincident graph pieces. Uniform parametrized convergence places the perturbed strand in a narrow middle substrip and makes it a crosscut between the two plugged ends. A simple crosscut separates the long sides even if it repeatedly backtracks. The approach just described cannot change from one side to the other without crossing this wall. Its sign is consequently the sign of that accessible face. The positive physical separation after the chart and node neighborhoods are fixed absorbs the $O(\delta)$ detour. Only near a stopping tip could the short approach continue around an untraced end; these are exactly the excluded neighborhoods. $\square$

8.4 A common weight for the distant wall

Lemma 8.7 (Changing a distant wall by a common weight). *Let C and W be nested lattice square contours, with a fixed positive macroscopic collar between them. The graph in this collar is the critical nearest-neighbor square lattice. Let T be a set strictly inside C , and let ν_1, ν_2 be two laws with plus spins prescribed on W . The laws may have different prescribed spins inside C , but have the same interactions and pinning rule on and outside C , where C is free. Write ν_i^b for the corresponding laws after replacing the plus wall by any assignment b on W .*

There is a positive function w_b of the spins on C , independent of the data strictly inside C , such that

$$d\nu_i^b|_{T \cup C} = \frac{w_b d\nu_i|_{T \cup C}}{\nu_i(w_b)}, \quad \frac{\sup w_b}{\inf w_b} \leq K.$$

The constant K depends only on the fixed collar geometry, uniformly in the mesh and in b . Consequently

$$\|\nu_1^b|_T - \nu_2^b|_T\|_{\text{TV}} \leq 2K \|\nu_1|_{T \cup C} - \nu_2|_{T \cup C}\|_{\text{TV}}.$$

Proof. For a contour assignment j , sum all spins strictly between C and W , keeping j on C and b on W . Denote this exterior partition weight by $A_b(j)$. Use the same allocation of energies along C in every law. The quotient

$$w_b(j) = \frac{A_b(j)}{A_+(j)}$$

does not involve any prescribed spin or interaction strictly inside C . Factoring the Gibbs density at C proves the stated common-weight identity, including its separate normalization for each ν_i .

Consider the Ising law on the collar alone with the spins on both contours free. Its joint marginal on (σ_C, σ_W) is proportional to $A_b(j)$. For any j, j' , the quantity

$$\log w_b(j) - \log w_b(j') = \log A_b(j) + \log A_+(j') - \log A_+(j) - \log A_b(j')$$

is therefore a log-odds rectangle. Lemmas 2.1 and 2.2 bound its absolute value by $4 \operatorname{arctanh} q_0$. The fixed square collar bounds this quantity by a geometric constant, exactly as in Corollary 2.5. Exponentiating proves the oscillation bound on w_b . Energies involving only wall spins multiply a partition weight by a scalar and hence do not affect this bound.

Rescale w_b so that $1 \leq w_b \leq K$. Put $d = \|\nu_1|_{T \cup C} - \nu_2|_{T \cup C}\|_{\text{TV}}$. For an event A of these joint spins, the layer-cake formula gives $|\nu_1(w_b \mathbf{1}_A) - \nu_2(w_b \mathbf{1}_A)| \leq Kd$ and $|\nu_1(w_b) - \nu_2(w_b)| \leq Kd$. Hence

$$\left| \frac{\nu_1(w_b \mathbf{1}_A)}{\nu_1(w_b)} - \frac{\nu_2(w_b \mathbf{1}_A)}{\nu_2(w_b)} \right| \leq 2Kd.$$

Take the supremum over A and then restrict to T . □

Application to the outward erosion lemma. It suffices first to prove Lemma 8.1 for a constant plus distant wall. Indeed, choose a square contour C between the entire compact target T and the distant wall W , with positive clearances from both and enclosing the full permitted stopping band. All three inner laws μ^0, μ^+, μ^f have the same free collar from C to W . Apply the constant-wall erosion construction to the enlarged compact target $T \cup C$, with its sum of errors smaller than $\varepsilon/(2K)$. The extra contour remains a positive distance from the stopping band and from W , so this target satisfies the same geometric hypotheses.

For any assignment b on W , Lemma 8.7 now transfers each of the two joint errors to T , and their sum is less than ε . The weight is the same for every transcript and for both insertions. Thus the fully inserted law is still determined only by the selected barrier, the mesh, and b , independently of the transcript. Adding plus and then minus pins retains the same two orders under this prescribed wall. Every interior transcript remains realizable under the plus reference wall, since finite Gibbs weights are positive.

The constants and the choice of the barrier are uniform in b ; no limiting boundary description for a mesh-dependent assignment is used. The marked-domain convergence argument above is therefore applied only with a constant distant wall. More generally its continuum mean assertion requires fixed finitely many nondegenerate signed or free arcs on every boundary component. Arbitrary lattice wall assignments enter through the common weight proved above.

8.5 Construction and proof of local erosion

Proof of Lemma 8.1. By the preceding common-weight reduction, take the distant wall constant in the outward case. Keep the source and test contours fixed as above. We construct the barrier and the two mean bounds used in Equation (8.3). The choices will be made in the order in which their uniformity is needed: node neighborhoods, comparison outlines, erosion contour, and finally the datum tolerance and lattice mesh.

Node neighborhoods and their mean error. Use the chart of Lemma 8.2. Choose disjoint small closed neighborhoods $P_1, \dots, P_k$ of the roots and tips. Include one auxiliary node on the base if needed to cut its circular strip into finitely many coordinate strips. Each neighborhood is the image of a small rectangle in the chart. Choose a smaller rectangle strictly inside each one, so that every incident ideal edge is monotone in its local strip coordinate until it leaves the larger rectangle. The incident edges use separate prescribed strips; the nested rectangles leave positive slack for the joins constructed below. Compact parts of nonincident arcs have positive separation.

In a *plugged* law, fix every lattice spin in these neighborhoods to the same assignment, in every law under comparison, overwriting any old prescription there. Denote this operation by a superscript P . These plugs will close the channels between displaced boundaries. Their assignments need not agree with the old side signs.

For a single plug and any of the laws λ considered below, Corollary 2.5 and the moment bound give

$$|\mathbb{E}_\lambda M_J^\delta - \mathbb{E}_{\lambda^P} M_J^\delta| \leq \omega(\text{diam } P), \quad \omega(t) \longrightarrow 0. \quad (8.8)$$

The estimate is uniform in all other pins and for sufficiently fine meshes. Indeed, the contracted connection probability in Lemma 2.2 is symmetric in its terminal sets. Its small-patch bound controls the density on the remote contour J^δ when the assignment in P changes. If that density ratio is R_P , then $e^{-b_P} \leq R_P \leq e^{b_P}$, where $b_P \rightarrow 0$ as the physical plug diameter tends to zero.

Hence

$$\left| \mathbb{E}_\lambda[(R_P - 1)M_J^\delta] \right| \leq (e^{b_P} - 1) \left(\mathbb{E}_\lambda(M_J^\delta)^2 \right)^{1/2}.$$

When some spins in P were free, first condition on all their values, apply the uniform assignment comparison, and average. To overwrite old pins, first remove their prescriptions while retaining all vertices and edges, and use the same conditional comparison. Pins outside P are common. No free collar is needed around the plug, whereas J retains its fixed free collar and therefore has a uniform moment bound. Sum (8.8) over the finitely many plugs. Choose their sizes so that the sum is less than a small number $a > 0$, for every law used in the proof. We will choose a according to the fixed transmission constant and the requested total-variation error.

Regular upper and lower boundaries. We next construct deterministic comparison laws indexed by a small chart displacement $q > 0$. On a one-sided base arc, move a plus face towards the fresh side and a minus face away from it by amounts comparable to q . On a slit, move a thin ribbon entirely towards the plus side by a distance comparable to q , keeping the respective plus and minus face signs. Its thickness is a small fixed fraction of q . These choices give the upper outline. Reverse both moves for the lower outline. Join the pieces near roots and tips using fixed scaled polygonal templates, with one sign-switch mark on each cap and no neck narrower than a fixed fraction of the template scale. For sufficiently small q , all joins lie in the smaller node rectangles. Regularly approximate each outline by Lemma 8.3, within a still smaller fraction of its clearances. Let ν_q^u and ν_q^l be the unmodified Gibbs laws, with the same distant wall in the annular case.

Lemma 8.4 makes their continuum one-point functions converge to the same limit as $q \downarrow 0$. For each fixed q , Theorem 1.3 of [4] gives convergence of the normalized lattice means uniformly on J . Since J^δ has $O(\delta^{-1})$ vertices,

$$\lim_{q \downarrow 0} \limsup_{\delta \downarrow 0} |\mathbb{E}_{\nu_q^u} M_J^\delta - \mathbb{E}_{\nu_q^l} M_J^\delta| = 0. \quad (8.9)$$

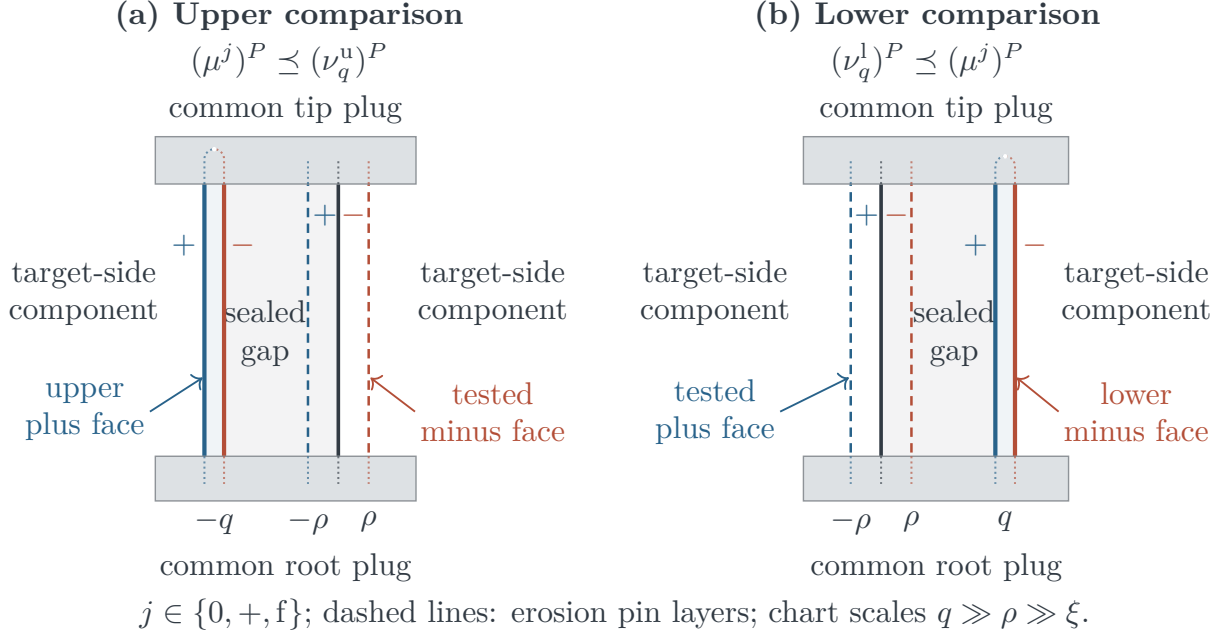
Choose q and its regular approximations so that this gap is small for all sufficiently fine meshes. This choice is made after the plug sizes, and keeps every join strictly inside its plug.

The erosion contour and its free neighborhood. Now choose $0 < \rho \ll q$. At chart distance ρ on the fresh side, put an outline parallel to the base and around both sides and the tip of each slit. Prescribe the face signs, with one switch on each cap. This is a Jordan contour. A sufficiently accurate regular approximation gives Γ_d . Choose its switches on grid-aligned straight portions. For fine mesh, a simple lattice cycle following it, with a boundary layer if necessary, has the prescribed cyclic arc order and separates the target from the old boundary.

After ρ and this approximation are fixed, the new layer has positive physical separation from the ideal datum. Choose the datum neighborhood and mesh small enough to preserve it. The recorded lattice datum uses the map R_δ from Section 7; undoing that map moves its curves by only $O(\delta)$ in the physical plane. Uniform continuity of the fixed chart and its inverse therefore turns a sufficiently small recorded-data tolerance, followed by a sufficiently fine mesh, into the required chart tolerance for the actual wall. Absorb this displacement and the examined triangles' $O(\delta)$ thickness in the remaining clearance. The entire new layer is then free in the original conditional law.

In the inward case the same choice preserves more than the layer. Let D be the Jordan base interior with its attached slits removed. The outline wraps around every slit, so $\text{Int } \overline{\Gamma_d} \subset D$. It has a thin exterior collar whose union with this closed interior is a compact set $C \Subset D$. Uniform parametrized convergence of the base preserves its winding number about every point of C , uniformly because C has positive distance from the base. The nearby slits stay away from C as well. Thus C lies in the accessible domain of every sufficiently close realized datum. Nonvisible traces

and hidden pockets lie in its blocked complement. All examined vertices are on the initial side or within $O(\delta)$ of the traced boundary. It follows that the entire component kept inside Γ_d , together with a positive exterior collar, is free of old pins, uniformly over the chosen datum neighborhood.



The same continuation for each ordered insertion

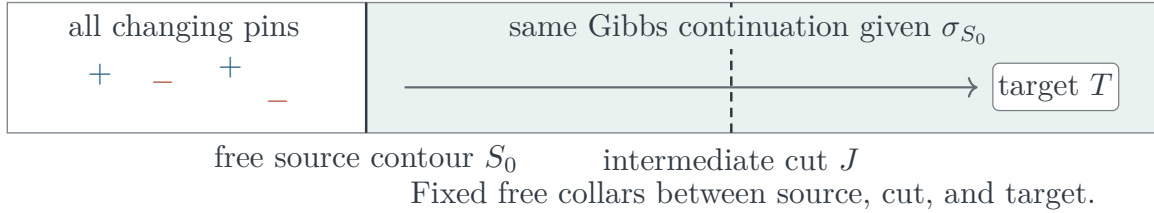


Figure 1: Ordered comparison in a topological slit chart, and the common continuation used for each insertion. The old slit has plus on its left and minus on its right; the black tested wall has chart coordinate zero. The displayed inequalities concern the laws restricted to the common target component. In (a) the upper ribbon moves towards the plus side; in (b) the lower ribbon moves towards the minus side. Common root and tip plugs seal each interwall channel. All ends extend into the plugs with slack; dotted cap traces show geometry overwritten by the common plug assignment. The common free target component sees an upper plus or tested minus face in (a), and a tested plus or lower minus face in (b), or identical plug data. The exterior sectors may communicate outside the chart. Dashed erosion layers may supply the tested frontier after insertion; the opposite layer is screened. The lower strip shows the shared source contour S_0 , an intermediate cut J , and the target T : all changed facing-boundary pins precede S_0 , and for each fixed distant wall the conditional continuation beyond it is the same. The coordinates are chart coordinates, with $q \gg \rho \gg \xi$; no metric regularity of the chart is assumed.

Incidence ordering for every nearby transcript. Let μ^0, μ^+, μ^f be the three laws in the statement. We compare their plugged versions to the same two plugged regular laws. Figure 1 shows the local geometry. First consider a slit strip outside the node plugs. Give the old slit plus on its left

and minus on its right, and put it at chart coordinate $x = 0$. The actual perturbed strand lies in $|x| < \xi$, where $\xi \ll \rho$. The upper ribbon is near $x = -q$, the erosion plus layer near $x = -\rho$, and the erosion minus layer near $x = \rho$.

Parametrized convergence also controls visits to the strip ends. For each ideal edge, its part between the smaller node rectangles lies in a strip separated from all nonincident edges. On the matched perturbed edge, select its last exit from the first smaller rectangle before its first entry into the second. This gives a simple crosscut. The ideal edge is monotone near each node, so closeness of the parametrizations places these exit and entry parameters near the corresponding ideal parameters. The positive gap between the smaller rectangle and the actual plug therefore keeps the entire discarded prefix and suffix inside their respective plugs. For example, in a local edge coordinate with smaller-rectangle exit at $s = 2h$ and plug exit at $s = 3h$, a tolerance $e < h/3$ puts every point of the discarded prefix at $s \leq 2h + 2e < 3h$.

Retain the entire selected crosscut, including any later visits to an outer plug. Every portion of the original edge outside the actual plugs is then retained. Turns and repeated plug visits do not change the crosscut's separation of its two long sides: filling the plugs can remove components but cannot join the two sides. Its orientation identifies the accessible face signs, as in Lemma 8.6. This argument applies to every slit and to each base arc between consecutive nodes. There are only finitely many edges, so one tolerance works for all of them. Taking the mesh below the remaining physical slack prevents a lattice edge from passing between a wall end and its plug.

The upper ribbon and this old-wall crosscut, joined through the wholly pinned root and tip plugs, close the channel between them. That channel cannot meet the target component of the sites free in both laws. Additional examined pieces cannot reopen it. The common target component can consequently see only the plus face of the upper ribbon or the minus face of the old wall. In the fully eroded tested law, the latter frontier may instead be the erosion minus layer. In the plus-only tested law, its extra plus pins lie in the screened channel. The same upper comparison therefore works for all three laws. Lemma 8.6 identifies each accessible examined-neighbor sign with the appropriate wall face; the recorded-data displacement was already absorbed into its physical slack. At plug incidences the signs agree by construction.

For a base arc the verification is one-sided. If its sign is plus, the upper outline is nearer the fresh side, so the upper model supplies a plus pin at the common component's frontier. If its sign is minus, the tested wall is nearer and supplies a minus pin. The sign-changing joins are inside the common plugs. Thus every incidence satisfies Lemma 8.5, and $(\mu^j)^P \preceq (\nu_q^u)^P$ on the common target component for $j \in \{0, +, f\}$.

For the lower comparison move the ribbon to the minus side. The accessible incidences are the lower minus face or the tested plus face. Additional tested plus pins reinforce this order; the erosion minus pins lie in the screened channel. The base-arc argument is reversed in the same way, and the plugs again give identical signs. Applying the incidence criterion proves the full sandwich

$$(\nu_q^l)^P \preceq (\mu^j)^P \preceq (\nu_q^u)^P, \quad j \in \{0, +, f\}. \quad (8.10)$$

Mean bounds and the completed law. Removing the plugs changes each mean by less than a . Thus (8.10) places every $\mathbb{E}_{\mu^j} M_j^\delta$ between the two unmodified regular means with error at most $2a$ at either end. Each of the two ordered mean differences in (8.3) is at most the absolute regular mean gap plus $4a$. Choose a first and then q so that Equation (8.9) and this error make the sum of target total-variation changes less than ε . All later choices of ρ , regularization, neighborhood, and mesh preserved these bounds. No spin-scaling limit in a realized transcript's irregular domain has been used.

Finally, the completed layer separates the target from every old examined vertex. Conditional

factorization of the Gibbs weights therefore leaves exactly the law with the chosen new boundary and the unchanged distant wall. It is independent of the transcript. The argument conditions on the full transcript, including the actual auxiliary radii when they are random. Omitted initial spins and nonvisible traces lie on the blocked side of the new barrier. A stopping radius producing no visible tip adds no boundary piece. Thus the same completed law works even when the datum does not record its numerical radii. Averaging over these radii or other omitted information preserves the two total-variation bounds. $\square$

Lemma 8.8 (Moving a completed barrier outward). *In the inward setting, let a regular signed contour Γ have already been fully pinned, and let V^δ be the entire free component it encloses, with its pinned boundary layer excluded. Assume that a positive exterior collar of Γ contains no other pins, as supplied by Lemma 8.1. For any compact target on its inner side and any $\varepsilon > 0$, one can choose a regular surrounding contour $\hat{\Gamma}$ at a positive distance from Γ , with corresponding signs, so that the following two steps on the marginal on V^δ each change the target law by less than ε :*

1. *remove the old plus pins on Γ ;*
2. *remove the old minus pins on Γ .*

Here the farther pins on $\hat{\Gamma}$ are present from the start. They initially have no effect on V^δ . The first step is downward and the second upward. After both, the kept volume V^δ is separated from every fixed pin by a positive macroscopic distance, uniform for sufficiently fine meshes.

Proof. Choose $\hat{\Gamma}$ as a sufficiently thin outward offset of Γ , matching the finite sign arcs. Put the new pins there before changing the old ones. The full old layer disconnects the kept volume, so these farther pins are initially harmless. We retain the same V^δ throughout the argument; in particular we do not first discard a free strip inside the old layer.

Use the construction of Lemma 8.1 without slits. Put small identical plugs around the finitely many sign switches, make upper and lower regular outlines by moving plus and minus arcs in their respective ordered directions, and compare by Lemma 8.5. The test contour lies strictly inside Γ , with a common free collar. As the offset tends to zero, both unmodified envelope means have the same limit by Lemma 8.4; the plug errors tend to zero by Corollary 2.5. This controls the means for the original law, the law with only the old minus pins remaining, and the law with all old pins removed. The two actual removals are ordered, so Corollary 4.12 proves the claimed target estimates.

Fix the sufficiently small offset after obtaining these estimates. Its physical distance from Γ is positive, and mesh rounding preserves a fixed fraction of it. The kept volume is inside Γ ; the effective boundary pins are on the farther contour, and the original pins are farther away, beyond the available collar. This proves the final buffer assertion. $\square$

9 Finite stopping-band descriptors

Local erosion assigns a barrier to a neighborhood of each admissible datum. We now choose finitely many of these neighborhoods covering all but a prescribed amount of the reference probability. This is a finite approximation in probability; the whole space of slit data need not be compact, and no deterministic bound on every possible slit count is required.

Theorem 9.1 (Stopping-band transfer). *Fix an inward or outward band, target, and ambient square chart as in Section 7, with all the stated positive clearances. Use the critical plus-boundary*

law in the fixed larger reference square. For every exceptional probability $\epsilon > 0$ and target error $\eta > 0$, generic contour levels can be chosen in any prescribed separated subwindows so that the following hold for all sufficiently fine meshes.

1. The stopping observation of Lemma 7.2 has a finite descriptor $Z^\delta \in \{0, 1, \dots, m\}$, computed from band spins. State 0 is exceptional and has probability at most ϵ in the fixed reference chart.
2. Every good state i has a deterministic regular signed barrier Γ_i . For every realizable transcript with $Z^\delta = i$, adding its plus pins and then its minus pins changes the target law by a sum of at most η in total variation. The resulting target-side Gibbs law, denoted $Q_{i,\delta}$, is the same for every such transcript. This assertion is uniform over all bounded observables on the target.
3. In the inward case one may also move the full barrier to a slightly farther signed contour, retaining the entire enclosed free component as the kept volume. Counting the two old-pin removals as well, the sum of target errors can still be made at most η . The final kept volume has a positive macroscopic buffer from its fixed boundary. All geometric clearances have positive minima over the good states.
4. For a fixed finite list of bands the descriptors have convergent joint laws. The bins can be chosen to be continuity sets of the limiting data law; hence this finite-space convergence is in total variation.

The number of states, their barriers, and their positive clearances depend on the fixed geometry and on ϵ, η , not on the mesh. The threshold for the mesh is chosen afterward. If a different law, possibly conditional on remote pins, has band density at most K times that of the reference chart by Corollary 2.5, its exceptional probability is at most $K\epsilon$. The same applies to mixtures of those conditional laws.

Proof. Choose generic levels by Lemma 7.4. At each admissible limiting datum apply Lemma 8.1, and, for the inward version, Lemma 8.8. Allocate the requested total-variation budget among the finitely many ordered steps. This produces an open neighborhood and a barrier construction valid uniformly for every realizable datum in that neighborhood on all sufficiently fine meshes.

Refine this cover to balls whose closures lie in such neighborhoods and whose boundaries have limiting probability zero. For a fixed center, only countably many radii can give positive boundary mass. Separability from Remark 7.5 supplies a countable such subcover. By continuity of probability along its increasing finite unions, take finitely many balls whose union has limiting probability larger than $1 - \epsilon/2$. Order them and form disjoint bins by assigning a datum to the first ball containing it. Their boundaries are contained in the finite union of the selected sphere boundaries and therefore have probability zero. The remaining bin is exceptional. Convergence of the data makes its lattice probability at most ϵ for all sufficiently fine meshes.

Associate to a good bin the barrier of the neighborhood containing its ball. The local lemmas hold for every transcript in that bin, with no dependence on an observable. Take the minimum of the finitely many positive clearances and mesh thresholds. This proves the first three assertions. For a finite list use the same argument in each coordinate and Lemma 7.4 jointly. Every descriptor string is a continuity set, proving convergence of all its finitely many probabilities. Finally the bad event depends only on band spins, so the density comparison gives the last assertion directly, including after conditioning on the stated remote information. $\square$

Corollary 9.2 (Lookup approximation of conditional averages). *In the outward setting of Theorem 9.1, let $\mathcal{F}^\delta$ be the stopping information. For any family of target observables h_δ satisfying $|h_\delta| \leq H$, define*

$$a_i(\delta) = \mathbb{E}_{Q_{i,\delta}} h_\delta \quad (i \geq 1),$$

and choose any exceptional-bin value in $[-H, H]$. Then, on every good transcript,

$$\left| \mathbb{E}[h_\delta \mid \mathcal{F}^\delta] - a_{Z^\delta}(\delta) \right| \leq 2H\eta. \quad (9.1)$$

In particular, in the reference chart,

$$\mathbb{E} \left| \mathbb{E}[h_\delta \mid \mathcal{F}^\delta] - a_{Z^\delta}(\delta) \right|^2 \leq 4H^2(\eta^2 + \epsilon). \quad (9.2)$$

The good bins, barriers, and mesh threshold work for all these observables simultaneously. Neither the observables nor their lookup values need converge as $\delta \downarrow 0$. More explicitly, fix a reference continuation law ν and regard $f_\delta^\nu(T) = \mathbb{E}_\nu[h_\delta \mid \mathcal{F}^\delta = T]$ as a function of the stopping transcript T . Evaluating this same function under any buffered band distribution with density bound K gives the integrated estimate with exceptional term $K\epsilon$. It is the reference conditional-average function that is transferred. It also equals the conditional average for the new law when that law has the same continuation and distant wall given T . If the distant wall is changed to an assignment b , the uniform wall extension in Lemma 8.1 supplies the corresponding completed laws $Q_{i,\delta}^b$. The conditional averages for this continuation use the separate lookup values $a_i^b(\delta) = \mathbb{E}_{Q_{i,\delta}^b} h_\delta$.

Proof. The conditional target law of a stopping transcript is its Gibbs continuation by Lemma 7.2. On a good bin its total variation distance from the single law $Q_{i,\delta}$ is at most η . The defining inequality for total variation gives Equation (9.1). On the exceptional bin both quantities are bounded by H in absolute value. Squaring and integrating proves Equation (9.2). All constructions in Theorem 9.1 preceded the choice of h_δ . The pointwise good-bin estimate and the transferred exceptional probability prove the last assertion. This is the reason that mesh-dependent likelihoods can be transferred without assigning them a continuum limit. $\square$

Corollary 9.3 (Random levels and finite ball probes). *Theorem 9.1 and Corollary 9.2 remain valid when the two contour radii are drawn with a bounded density in a fixed compact rectangle of separated radii lying strictly inside the permitted windows. Different band observations may use independent draws. Their joint data laws converge, including the auxiliary randomness.*

Fix a countable dense collection of marked rational-walk prototypes in the ambient space $\hat{\mathcal{X}}$ of Definition 7.3; probe balls are intersected with the admissible-data space $\mathcal{X}_{\text{band}}$. Allow a probe to test whether the data lie within distance r of one prototype, where r is drawn uniformly in a rational interval with closure in $(0, 1)$. For any prescribed ϵ, η , finitely many such probes suffice to obtain the lookup approximation in Corollary 9.2, with adjusted exceptional probability at most ϵ . The finite probe list depends only on the reference local geometry, the radius rectangle, and the requested tolerances. It is independent of the bounded observables and of the size of an ambient buffered array.

Proof. For almost every radius pair the deterministic continuity argument in Lemma 7.4 applies. Couple the radius draws identically on all meshes, independently of the converging loops. The records take values in the union over the permitted radii specified in Definition 7.3. The deterministic rounding rule is measurable in those draws, and its recording maps R_δ have uniformly vanishing displacement. Thus the continuous mapping theorem, or dominated convergence after this coupling,

proves joint convergence of the encoded data and the radius draws. The full stopping information includes those draws. The compact radius rectangle provides a uniform positive separation of contours and uniform clearance from the window edges. Thus the local erosion construction still has a neighborhood for every admissible resulting datum.

For each datum choose a neighborhood U_d as in Lemma 8.1. By density, there are a prototype p and rational numbers $0 < a < b < 1$ such that

$$d \in B(p, a), \quad B(p, b) \subset U_d.$$

Choose a rational threshold interval strictly between a and b . A positive test then certifies membership in U_d , while every datum in $B(p, a)$ tests positive for every threshold in that interval. The inner balls form an open cover of the admissible data. Select finitely many covering probability at least $1 - \epsilon/2$ in the reference law, and decode by taking the first positive test in their fixed order. If no test is positive, use the exceptional state. For a positive test use the single barrier law belonging to its neighborhood U_d ; hence Equation (9.1) holds without dependence on the random threshold or on h_δ .

Every probe tie has probability zero because its threshold is an independent continuous random variable, conditional on the data. The finite string of bits consequently converges in law, and the exceptional probability is at most ϵ for fine meshes. Only finitely many entries in any fixed dense enumeration of the prototypes and rational intervals have been used. For finitely many permissible radius windows, take the union of the corresponding lists. Finally the choices were made in one fixed local reference chart. Density comparison transfers their error bounds to other buffered chart laws with the same fixed multiplicative constant, regardless of the number of tiles in the larger array. $\square$

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